

A Perspective-and-Aspect-Driven System of Temporal Anchoring in Portuguese: Semantic Reanalysis of the Present Perfect and Other Cross-Romance Divergences¹

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Abstract:

This theoretical paper — building on and developing the author's earlier Japanese works (Makino 2010, 2012, 2013) — proposes a feature-based analysis of the indicative and subjunctive tenses in Portuguese, grounded in four primitive anchoring-aspect features and their figurative extensions: [non-assertive], [VP_{past}], [finished] and [assumptive]. In contrast to the tense-driven systems typical of other Romance languages, Portuguese is assumed to employ a perspective-and-aspect-driven system. Like the former systems, it takes the speech time as the anchoring point. Portuguese, however, instead of establishing reference times such as [±past], divides the time axis into two intervals: one extending backward and excluding the speech time — the PAST — and the other extending forward and including it — the NON-PAST. Simultaneously, the speaker's temporal viewpoint (VP) is located at some point either within the NON-PAST — by default, at the speech time — or within the PAST. From this VP, the situation to be described is evaluated as either finished or not finished. Its temporal anchoring is thus secondarily derived from the location of the VP and from whether the situation is viewed as finished or not from that VP. This structural opposition between the two systems accounts for a range of cross-Romance differences, including the uniquely Portuguese reanalysis of the present perfect indicative (PPC), the functional erosion of the future tense and the associated future-oriented use of the preterite (PPS), and the remarkable vitality of the PPS in this language.

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¹ **How to cite:** Makino, S. (2025). *A Perspective-and-Aspect-Driven System of Temporal Anchoring in Portuguese: semantic reanalysis of the Present Perfect and other Cross-Romance divergences*. Preprint, LingBuzz/Zenodo. DOI [10.5281/zenodo.15420846](https://doi.org/10.5281/zenodo.15420846).

0. Preface to the Preprint Edition

The semantic reanalysis of the Portuguese present perfect indicative (*pretérito perfeito composto do indicativo*, PPC) stands out among the various cross-Romance differences accounted for by the opposition between the two temporal anchoring systems proposed in this paper. This preprint (v9) builds on existing research while introducing a more unified framework for explaining this reanalysis — a framework grounded in feature-based structure, paradigmatic saturation, functional competition and redundancy avoidance, as well as in aspectual reinterpretation, feature reconfiguration and cognitive-frame emergence.

Rather than attributing this semantic change to morphosyntactic restructuring or lexical drift, the account locates its source in deeper structural factors. At the time when the PPC emerged, Portuguese's core temporal-aspectual system — defined by a 2×2 feature grid of $[\pm VP \in \text{past}] \times [\pm \text{finished}]$ or $[\pm \text{past}] \times [\pm \text{finished}]$ (cf. Section 2.1.2) — was already fully saturated by the *present*, *preterite*, *imperfect* and *pluperfect* indicatives. Given that this core paradigm — which appears to have remained aspectually unchanged since Classical Latin, in that it **formally encoded the aspectual feature $[\pm \text{finished}]$** , comprising the quasi-equivalents of *praesens*, *perfectum*, *imperfectum* and *plusquamperfectum* — was already exhaustively saturated, the introduction of any additional form would have resulted in redundancy.

Originally, PPC₁ encoded $[-VP \in \text{past}, +\text{finished}, [\checkmark \text{initiated}, \checkmark \text{result-stative}, -\text{finished}]]$, thereby directly overlapping with the simple preterite (*pretérito perfeito simples do indicativo*, PPS), specified as $[-VP \in \text{past}, +\text{finished}]$. In order to remain viable, PPC₁ relinquished the feature $[+\text{finished}]$, which placed it in direct competition with the PPS. And, through a synecdochic reinterpretation of its result state $[\checkmark \text{initiated}, \checkmark \text{result-stative}, -\text{finished}]$ — which, crucially, the PPS could not denote independently without adverbials (e.g. *já*, *ainda não*) or other contextual or/and situational cues, and which was the only oppositive element between the two — into $[\checkmark \text{initiated}, \checkmark \text{stative}, -\text{finished}]$, it became PPC₂:

PPC₁ → PPC₂

$[-VP \in \text{past}, +\text{finished}, [\checkmark \text{initiated}, \checkmark \text{result-stative}, -\text{finished}]]$

↓

$[-VP \in \text{past}, +\text{finished}, [\checkmark \text{initiated}, \checkmark \text{result-stative}, -\text{finished}]]$

↓

$[-VP \in \text{past}, \checkmark \text{initiated}, \checkmark \text{stative}, -\text{finished}]$

The historical development from the earliest result-stative/possessive construction — "*haver* in the presente indicative + NP + past participle of a transitive verb" (e.g. *Hei um livro lido* "I have a book read") — to PPC₂ can be traced through **a series of structural reductions in both lexical and functional features**.

Initially, the construction consisted of the following constituents — here, "L" or "F" indicates whether a given feature is lexical or functional, [$\sqrt{\text{theme}}$] globally represents the theta-role assigned by a transitive verb and "pp." is an abbreviation of past participle:

- *haver*: L [$\sqrt{\text{possessive}}$] / F [$-\text{VP} \in \text{past}$, $-\text{finished}$]
- pp.: F [$+\text{finished}$] [$\sqrt{\text{initiated}}$, $\sqrt{\text{result-stative}}$]
- NP: L [$\sqrt{\text{theme}}$, $\sqrt{\text{possessed}}$]

When [$\sqrt{\text{possessive}}$] and [$\sqrt{\text{possessed}}$] were removed from *haver* and the NP, respectively, the structure became as follows:

- *haver*: F [$-\text{VP} \in \text{past}$, $-\text{finished}$]
- pp.: F [$+\text{finished}$] [$\sqrt{\text{initiated}}$, $\sqrt{\text{result-stative}}$]
- NP: L [$\sqrt{\text{theme}}$]

When the construction was extended to intransitive verbs, the transitive relation between the verb and its argument NP was no longer required, resulting in the following configuration — all the remaining features were functional and the notation F has therefore been omitted:

- *haver*: [$-\text{VP} \in \text{past}$, $-\text{finished}$]
- pp.: [$+\text{finished}$] [$\sqrt{\text{initiated}}$, $\sqrt{\text{result-stative}}$]

Although *haver* was replaced by *ter*, the structural configuration remained the same — PPC₁:

- *ter*: [$-\text{VP} \in \text{past}$, $-\text{finished}$]
- pp.: [$+\text{finished}$] [$\sqrt{\text{initiated}}$, $\sqrt{\text{result-stative}}$]

Finally, when [$+\text{finished}$] was deleted and, as a necessary consequence, the [result-] component was removed from [$\sqrt{\text{result-stative}}$] via synecdoche, the structure resulted in PPC₂:

- *ter*: [$-\text{VP} \in \text{past}$, $-\text{finished}$]
- pp.: [$\sqrt{\text{initiated}}$, $\sqrt{\text{stative}}$]

This is assumed to constitute the historical pathway from the earliest result-stative/possessive construction to PPC₂. **Ultimately, the enigma of the Portuguese present perfect indicative lies in the semantic reanalysis of the participle itself — from a [+finished [√initiated, √result-stative]] participle in PPC₁ to a [√initiated, √stative] participle in PPC₂ — due to its competition with the PPS over [+finished], which is thus the key concept in the semantic reanalysis of the PPC.** By contrast, the features [−VPE_{past}, −finished] on the auxiliary *haver/ter* remained virtually unchanged throughout. Accordingly, all variation in the meanings of perfect constructions can, first and foremost, be traced back to this strikingly singular shift in participial semantics. In fact, in Portuguese, in all compound tense forms except the PPC, the past participle was reanalysed — by deleting the [√initiated, √result-stative] bundle — as [+finished], with the simultaneous abandonment of the [−finished] value in the auxiliary.

Based on this feature reconfiguration, the new feature [√stative] in PPC₂ came to coerce an iterative-stative interpretation of eventive situations. An event consists of a sequence of distinct phases and therefore does not permit homogeneous continuity: in other words, unlike a state, a part does not equal the whole. To homogenise — that is, to stativise — it, one only needs to repeat it. This is because any portion of a repetition still counts as a repetition.

Through habitual use, this interpretation gave rise to a cognitive frame — "the initiation and continuation of an iterative state from a certain point in the past up to the speech time" — which, once associated with PPC₂, began to trigger an iterative-stative reading even for inherently stative predicates. When the context — including adverbials, the *Aktionsart* of the predicate and encyclopedic knowledge — fails to align with this frame, the frame is neutralised, yielding instead a simple continuous-stative interpretation.

PPC₂ is, furthermore, incompatible with temporal adverbials that explicitly quantify the number of iterations. This is because such quantification presupposes a time interval that is closed at both ends: counting events requires not only a defined starting point but also a defined endpoint. By contrast, PPC₂ denotes a time interval that is open at the endpoint due to the feature [−finished], thereby resulting in a semantic mismatch.

In sum, PPC evolves from PPC₁ [−VPE_{past}, +finished [√initiated, √result-stative, −finished]] to PPC₂ [−VPE_{past}, √initiated, √stative, −finished], acquiring a unique profile among Romance perfects by ceasing to encode finished events with present relevance. Why did the compound perfect form not replace the simple preterite form in Portuguese, as it did in other Romance languages? The answer lies in the fact that **the form that appeared to be a *preterite* — the**

PPS — was, in fact, a *perfect* form, with "perfect" here denoting finishedness without reference to a result state, as in the Classical Latin *perfectum* in its primary sense.

While previous studies have offered valuable insights, this paper builds upon them in an attempt to provide a more integrated explanatory model. The author welcomes scholarly feedback and dialogue prior to peer-review submission.

1. Introduction and Theoretical Orientation

Excluding the imperative mood, which does not exhibit any temporal-reference opposition, Portuguese finite verb forms can be classified into the combinations of mood and so-called tense in (1)². Note that, in this paper, the term "perfect" is used to designate any compound form consisting of an auxiliary verb plus a past participle, with reference solely to its morphological structure. In order to make formal correspondences explicit, a unified terminological system in English is employed for all temporal-reference forms in Portuguese, Spanish and French.

(1)

Indicative:	Present	Preterite	Imperfect	Pluperfect
	<u>Present Perfect</u>			
	Future	Future Perfect	Past Future	Past Future Perfect
Subjunctive:	Present	Present Perfect	Imperfect	Pluperfect

	Future	Future Perfect	_____	_____

(2) exemplifies (1), using the verb *falar* "to speak" in its third-person singular forms:

(2)

Indicative:	falo	falei	falava	tinha falado
	<u>tem falado</u>			
	falarei	tere falado	falaria	teria falado
Subjunctive:	fale	tenha falado	falasse	tivesse falado

	falar	tiver falado	_____	_____

As shown in (1) and (2), the subjunctive mood has no form that corresponds SEMANTICALLY to the present perfect indicative *tem falado*. The present perfect subjunctive *tenha falado* corresponds formally to *tem falado*, but semantically to the preterite indicative *falou*:

² The corresponding Portuguese names for these moods-tenses are, in the indicative mood, *Presente*, *Pretérito Perfeito Simples*, *Pretérito Imperfeito*, *Pretérito Mais-que-Perfeito*, *Pretérito Perfeito Composto*, *Futuro*, and *Futuro Composto do Indicativo*, along with *Condicional* and *Condicional Composto*; and in the subjunctive mood, *Presente*, *Pretérito Perfeito Composto*, *Pretérito Imperfeito*, *Pretérito Mais-que-Perfeito*, *Futuro*, and *Futuro Composto do Conjuntivo*. It should be noted, however, that the English terminology for Portuguese tenses is notoriously inconsistent, with a single tense often being referred to by multiple names. For example, the *Pretérito Perfeito Simples* may be translated as *Perfect*, *Past*, *Simple Past*, *Preterite*, or *Simple Preterite*.

(3) *A Teresa falou com o Pedro sobre isso.* "Teresa talked to Pedro about that."

(4) *Talvez a Teresa tenha falado com o Pedro sobre isso.* "Maybe Teresa talked to Pedro about that."

Both (3) and (4) have the same temporal-aspectual reference. Moreover, neither a past future subjunctive nor a past future perfect subjunctive exists. In this paper, we shall analyse the semantic oppositions among these forms, taking as our point of departure the opposition — assumed to be primitive — between the present *falo* and the preterite *falei* in the indicative.

The analysis in Section 2 reveals that each tense form corresponds to a specific combination of four primitive anchoring-aspect features, with semantic extension applied through conceptual metaphor, synecdoche and metonymy. These four features are:

- **mood:** [$\pm$ non-assertive] — suspension of judgment/commitment concerning the truth value of a sentence
- **perspective:** [$\pm$ VP $\in$ past] — see below
- **aspect:** [$\pm$ finished] — finishedness of a situation as viewed from the speaker's temporal viewpoint
- **modality:** [$\pm$ assumptive] — the speaker's belief in the possibility of the occurrence of a situation

In [VP $\in$ past], "VP" stands for the speaker's viewpoint, with [+VP $\in$ past] indicating that the speaker's viewpoint is located at some point in the past interval, and [–VP $\in$ past] indicating that the viewpoint is located at some point in the non-past interval — by default, at the speech time. In this way, each feature is specified either positively [+] or negatively [–]. Furthermore, it can be shown that the negative value [–] may assume a neutral value [0] when licensed by a specific context or situation.

The analysis also shows that two highly marked combinations — namely, [+non-assertive, +VP $\in$ past, +finished, +assumptive] and [+non-assertive, +VP $\in$ past, –finished, +assumptive], corresponding to the theoretically possible "past future perfect subjunctive" and "past future subjunctive" — are absent from the system. This corresponds to the cross-linguistically attested tendency to avoid overly marked grammatical forms. Additionally, provided that the temporal viewpoint is located exclusively on the speech time, an exceptional opposition marked by the **aspectual feature** [initiated] and the *Aktionsart* feature [stative] is observed only within the least marked area of the system — namely, [–non-assertive, –VP $\in$ past, –finished, –assumptive]

— and corresponds to the present indicative, specified as [0 initiated, 0 stative], and the present perfect, specified as [✓initiated, ✓stative].

The combination of the four primitive features — [non-assertive], [VP∈past], [finished] and [assumptive] — each taking either a positive or a negative value, logically yields 16 finite verb forms. Of these, two highly marked forms are excluded and one additional form is added, resulting in a total of 15 actual forms. These are exemplified in (5), using the verb *falar*'s third-person singular forms. The least marked area and the most marked one are shaded and the areas not filled with actual forms are indicated with strikethrough:

(5)

	[−finished]	[+finished]	[−finished]	[+finished]	
[−VP∈past]	[0 initiated, 0 stative] fala	falou	falará	terá falado	[−non-assertive]
	[✓initiated, ✓stative] tem falado				
[+VP∈past]	falava	tinha falado	falaria	teria falado	[+non-assertive]
[−VP∈past]	fale	tenha falado	falar	tiver falado	
[+VP∈past]	falasse	tivesse falado	————	————	
	[−assumptive]		[+assumptive]		

This system differs significantly, despite superficial similarities, from that of Reichenbach and its theoretical successor, the tense system proposed by Klein. In the latter frameworks, the temporal system of a language with absolute tense is wholly anchored in the speech time and the temporal location of a situation — state or event — is determined with respect to this absolute point of reference. Even among the Romance languages, Spanish — along with French and Italian — is one such language.

The finite verb forms in literary written Spanish may be classified according to the mood-tense combinations shown in (6), which are exemplified in (7) using the verb *hablar* "to speak" — the Spanish equivalent of Portuguese *falar* — in its third-person singular forms³. The three shaded mood-tense combinations, along with their corresponding example forms, are rarely used in modern spoken and written Spanish:

³ The corresponding Spanish names for these moods-tenses are, in the indicative mood, *Presente*, *Pretérito Perfecto Compuesto*, *Pretérito Imperfecto*, *Pretérito Pluscuamperfecto*, *Pretérito Perfecto Simple*, *Pretérito Anterior*, *Futuro Simple*, and *Futuro Compuesto del Indicativo*, along with *Condicional Simple* and *Condicional Compuesto*; and in the subjunctive mood, *Presente*, *Pretérito Perfecto Compuesto*, *Pretérito Imperfecto*, *Pretérito Pluscuamperfecto*, *Futuro Simple*, and *Futuro Compuesto del Subjuntivo*. We do not translate *pretérito anterior* as "preterite perfect", but rather as "preterite anterior", to avoid confusion with *pretérito perfecto simple*.

(6)	Indicative:	Present	Present Perfect	Imperfect	Pluperfect
		Future	Future Perfect	Preterite Past Future	<i>Preterite Anterior</i> Past Future Perfect
	Subjunctive:	Present	Present Perfect	Imperfect	Pluperfect
		<i>Future</i>	<i>Future Perfect</i>	_____	_____

(7)	Indicative:	habla	ha hablado	hablaba	había hablado
		hablará	habrá hablado	habló hablaría	<i>hubo hablado</i> habría hablado
	Subjunctive:	hable	haya hablado	hablara	hubiera hablado
		<i>hablare</i>	<i>hubiere hablado</i>	_____	_____

Its system of temporal anchoring first determines whether the situation to be described belongs to the actual world or to a non-actual one — a distinction encoded by the **mood feature** [$\pm$ non-assertive]. Next, starting from the speech time, two reference times are established relative to it: the past, which precedes the speech time and the non-past, which coincides with it. These two reference times are linguistically encoded by the **absolute tense feature** [$\pm$ past]. For each of these two primary reference times, two secondary reference times are then defined: the relative future, which is posterior to the primary reference time — encoded by either [$-$ past] or [$+$ past] — and the relative non-future, which is simultaneous with it. This opposition is encoded by the **relative tense feature** [$\pm$ posterior], which in Spanish is metonymically derived from the **modal feature** [$\pm$ assumptive]. With reference to one of the four reference-time combinations thus established — namely, "present" [$-$ past, $-$ assumptive], "future" [$-$ past, $+$ assumptive], "past" [$+$ past, $-$ assumptive] and "past future" [$+$ past, $+$ assumptive] — the situation time is evaluated as being either prior to it or simultaneous with it. This evaluation is encoded by the **situation-time feature** [$\pm$ anterior].

Finally, only in the case of situations marked as [$-$ non-assertive, $+$ past, $-$ assumptive, $-$ anterior] — and, in literary written Spanish, also those marked as [$-$ non-assertive, $+$ past, $-$ assumptive, $+$ anterior] — does the speaker mark two alternative locations for the speaker's temporal viewpoint, depending on whether it is located on the non-past reference time — namely, the speech time — or on the past reference time. This opposition is encoded by the **perspective feature** [$\pm$ VP $\in$ past].

We shall refer to this type of temporal system as a **tense-driven temporal system**. The

combination of the four major primitive anchoring-aspect features — [non-assertive], [past], [assumptive] and [anterior] — each taking either a positive or a negative value, logically yields 16 finite verb forms. Of these, two extremely highly marked forms are excluded and two additional forms — introduced by the feature [±VP∈past] — are added, resulting in a total of 16 actual forms that can be found in literary written Spanish. These are exemplified in (8), using the verb *hablar* in its third-person singular forms. The least marked area and the most marked one are shaded; theoretically possible areas not filled with actual forms are indicated with strikethrough; and forms rarely used in modern standard spoken and written Spanish are shown in italics:

(8)	[−assumptive]		[+assumptive]		
[−past]	hablo	ha hablado	hablará	habrá hablado	[−non-assertive]
[+past]	[+VP∈past] hablaba	[+VP∈past] había hablado	hablaría	habría hablado	
	[−VP∈past] habló	[−VP∈past] <i>hubo hablado</i>			
[−past]	hable	haya hablado	<i>hablare</i>	<i>hubiere hablado</i>	[+non-assertive]
[+past]	hablara	hubiera hablado	————	————	
	[−anterior]	[+anterior]	[−anterior]	[+anterior]	

(9) is identical to (8), except that the italicised forms rarely used in modern standard spoken and written Spanish have been excluded:

(9)	[−assumptive]		[+assumptive]		
[−past]	hablo	ha hablado	hablará	habrá hablado	[−non-assertive]
[+past]	[+VPEpast] hablaba	había hablado	hablaría	habría hablado	
	[−VPEpast] habló				
[−past]	hable	haya hablado	————	————	[+non-assertive]
[+past]	hablara	hubiera hablado	————	————	
	[−anterior]	[+anterior]	[−anterior]	[+anterior]	

We shall now examine the French temporal reference system. The finite verb forms in literary written French are classified according to the mood-tense combinations shown in (10), which are exemplified in (11) using the verb *parler* "to speak" — the French equivalent of Portuguese *falar* and Spanish *hablar* — in its third-person singular forms⁴:

⁴ The corresponding French names for these moods-tenses are, in the indicative mood, *Présent*, *Passé Composé*, *Imparfait*, *Plus-que-Parfait*, *Passé Simple*, *Passé Antérieur*, *Future Simple*, and *Future Antérieur du Indicatif*, along with *Conditionnel Présent* and *Conditionnel Passé*; and in the subjunctive

(10)	Indicative:	Present	Present Perfect	Imperfect	Pluperfect
		Future	Future Perfect	Preterite Past Future	Preterite Anterior Past Future Perfect
	Subjunctive:	Present	Present Perfect	Imperfect	Pluperfect
		_____	_____	_____	_____

(11)	Indicative:	il parle	il a parlé	il parlait	il avait parlé
		il parlera	il aura parlé	il parla il parlerait	il eut parlé il aurait parlé
	Subjunctive:	il parle	il ait parlé	il parlât	il eût parlé
		_____	_____	_____	_____

In this language, the modal feature [assumptive] — which is employed in Spanish — is replaced by the relative-tense feature [posterior], as shown in (12). Two forms in italics — namely, [+non-assertive, +past, −posterior, −anterior] and [+non-assertive, +past, −posterior, +anterior], corresponding respectively to the imperfect and pluperfect subjunctives — are rarely used even in this register today:

(12)	[−posterior]		[+posterior]		
[−past]	il parle	il a parlé	il parlera	il aura parlé	[−non-assertive]
[+past]	[+VP∈past] il parlait	[+VP∈past] il avait parlé	il parlerait	il aurait parlé	
	[−VP∈past] il parla	[−VP∈past] il eut parlé			
[−past]	il parle	il ait parlé	_____	_____	[+non-assertive]
[+past]	<i>il parlât</i>	<i>il eût parlé</i>	_____	_____	
	[−anterior]	[+anterior]	[−anterior]	[+anterior]	

In modern spoken French, however, the system in (12) is drastically reduced to that in (13), due to the deletion of the highly marked two forms specified as [−VP∈past] and the neutralisation of the [±past] opposition in the subjunctive area marked as [+non-assertive]. **Interestingly, in both Spanish and French, as in Portuguese, the most highly marked cell — in terms of feature values — remains unsaturated:**

mood, *Présent, Passé, Imparfait, Plus-que-Parfait du Subjonctif*. We do not translate *passé antérieur* as "preterite perfect", but rather as "preterite anterior", in order to make explicit the formal and structural correspondence with the Spanish *pretérito anterior*, which is likewise translated in this paper as "preterite anterior".

(13)	[−posterior]		[+posterior]		
[−past]	il parle	il a parlé	il parlera	il aura parlé	[−non-assertive]
[+past]	il parlait	il avait parlé	il parlerait	il aurait parlé	
[0past]	il parle	il ait parlé	————	————	[+non-assertive]
	[−anterior]	[+anterior]	[−anterior]	[+anterior]	

Portuguese diverges fundamentally from the aforementioned tense-driven system in that its temporal anchoring operates in a different way. Here — as in Spanish and French — after the mood feature [$\pm$ non-assertive] is specified to indicate whether the situation to be described belongs to the actual world or to a non-actual one, the speech time is taken as the primary anchoring point. Portuguese, however, instead of establishing reference times such as [$\pm$ past] on the basis of temporal relations to the speech time, divides the time axis, with reference to that point, into two intervals: one extending backward and excluding the speech time — the **PAST** — and the other extending forward and including the speech time — the **NON-PAST**. Simultaneously, the **speaker's temporal viewpoint for describing a situation** is located either within the NON-PAST — whose **default value is the speech time** — or within the PAST. From this located viewpoint, the situation to be described is then evaluated as either **finished** or **not finished**. In this system, what is crucial is whether the temporal viewpoint is located within the NON-PAST interval or within the PAST interval and whether the situation is viewed as not finished or finished. This is because the location of the temporal viewpoint linguistically encodes the opposition between two distinct worlds — the NON-PAST and the PAST — in terms of the possibility or impossibility of human intervention in the world as such, and the distinction between whether the situation is confirmed as not finished or as finished likewise encodes this same type of possibility or impossibility. **The temporal anchoring of a situation is secondarily derived from the position of the speaker's temporal viewpoint along the time axis and from whether the situation is viewed as finished or not from that viewpoint.** Crucially, in this model, tense is not treated as a primitive category. Rather, it is analysed as a derived grammatical value determined by the interaction of these primitive anchoring-aspect features: [$\pm$ VP $\in$ past] and [$\pm$ finished]. This perspective stands in contrast to traditional accounts, such as those of Reichenbach and Klein, which take tense — especially past, present and future — as the foundational basis of temporal interpretation. This may be called a **perspective-and-aspect-driven temporal system**.

The differences between the temporal anchoring processes in these two temporal systems are summarised in (14):

(14)

Temporal anchoring process				
	1 st stage	2 nd stage	3 rd stage	4 th stage
PT	[±VPEpast] speaker's temporal viewpoint located	[±finished] finishedness evaluated → situation time located	[±assumptive] occurrence possibility evaluated	
SP	[±past] reference times defined	[±posterior]←[±assumptive] sub-reference times defined	[±anterior] situation time located	[±VPEpast] viewpoint selected
FR	[±past] reference times defined	[±posterior] sub-reference times defined	[±anterior] situation time located	[±VPEpast] viewpoint selected

PT, SP and FR stand for Portuguese, Spanish and French, respectively.

In Portuguese, the temporal anchoring of a situation is processed in two stages: first, by applying the feature [±VPEpast] and then [±finished] — that is, [±VPEpast] ⇒ [±finished], where "⇒" indicates a transition from one stage to the next. Once the situation has been localised in future time — through the combination of the feature value [−VPEpast, −finished] and contextual cues such as adverbial expressions — the feature [+assumptive] is subsequently used to express the speaker's evaluation of the likelihood of the future event's occurrence. As such, this feature does not serve a temporal anchoring function. By contrast, French and Spanish require three — at most, four — stages in the temporal anchoring of a situation: [±past] ⇒ [±posterior] ⇒ [±anterior] (⇒ [±VPEpast]), or alternatively, [±past] ⇒ [±posterior] (← [±assumptive]) ⇒ [±anterior] (⇒ [±VPEpast]), where "←" indicates a figurative extension to another semantic domain. In both cases, the features [±posterior] or [±assumptive] are necessary for constructing a sub-reference time, which enables the anchoring of the situation in future time.

This fundamental opposition between these two systems helps clarify a number of differences observed between Portuguese and other major Romance languages, including Spanish and French. Why does the present perfect (*pretérito perfeito composto*) in Portuguese exhibit a temporal reference that is very different from that of the corresponding tense forms in the other languages? Why does the preterite use of the present perfect — widely recognised in French and partly in Spanish — not exist in Portuguese? Why has the range of uses of the preterite expanded in Portuguese, while in French the same form has long become obsolete in spoken language, with Spanish again occupying a transitional position? Why is it that the preterite (*pretérito perfeito simples*), when accompanied by the temporal adverbs *já* / *ainda não*, can be used to express future perfect meaning only in Portuguese? Why, in parallel, has the system of future tenses in Portuguese undergone obvious functional erosion, while in French the temporal localisation of future situations predominantly relies on the future tenses and Spanish appears to occupy an intermediate stage? Why does the choice of tense in counterfactual constructions

diverge in Portuguese from the patterns found in the other Romance languages? Why is the opposition between the preterite and the imperfect maintained in the subjunctive in Portuguese, whereas it is neutralised in the subjunctive in Spanish?

We shall account for such differences solely on the basis of **(a) the inventory of primitive anchoring-aspect features employed in each language; (b) the sequencing of those features within the anchoring process; and (c) the types of figurative extension applicable to those features.**

2. Primitive Semantic Oppositions among Finite Verb Forms in Portuguese

In this chapter, we shall extract the primitive features that constitute the system of temporal anchoring in Portuguese — the theoretical outline of which was presented in Section 1 — and examine their oppositional structure.

The present indicative bears overt morphological marking only for person and number, lacking any overt markers of mood or temporal reference. This can be seen by comparing the first-person plural form *fala-0-mos* of the verb *falar* in the present indicative with its counterparts in the imperfect indicative (*falá-va-mos*), future indicative (*fala-re-mos*), past future indicative (*fala-ria-mos*), present subjunctive (*fal-e-mos*), imperfect subjunctive (*falá-sse-mos*) and future subjunctive (*fala-r-mos*). While all of the latter forms bear overt markers of mood and temporal reference — namely, *-va*, *-re*, *-ria*, *-e*, *-sse* and *-r* — the present indicative lacks such markers, which is indicated here by *-0*. In this respect, **the present indicative** may be regarded as **a fully unmarked form** within the system of mood and temporal reference.

Just as the phonologically zero singular marker *-0* (e.g. *cat-0*) in English acquires its semantic value only through opposition to the overt plural marker *-s* (e.g. *cat-s*), the phonological zero of the present indicative — namely, *-0* — derives its values of mood and temporal reference only through opposition to other phonologically overt markers. This means that, by identifying the form that stands in minimal semantic opposition to the present indicative, we can isolate the minimal semantic value shared by the present indicative and that form. As seen in Section 2.1, the preterite indicative stands in opposition to the present indicative solely in that the former indicates the situation as *finished*, while the latter indicates it as *not finished*, thereby forming a minimal semantic pair. Moreover, this opposition encodes a principle that is empirically significant for human ontological cognition. On this basis, the present study takes the opposition between these finite verb forms to be the most primitive one and from this starting point seeks

to explore the semantic oppositions that hold among the other finite verb forms described above.

2.1 Fact-Marking Tense Forms: Present, Preterite, Imperfect and Pluperfect Indicatives

In Portuguese, the only verbal forms that present a situation as factual — apart from the present perfect indicative, which has a distinct temporal interpretation — are the present, preterite, imperfect and pluperfect indicatives. Using the third-person singular of the verb *falar*, these correspond to *fala*, *falou*, *falava* and *tinha falado*, respectively. As we shall see later, these four forms stand in the following temporal relations: *fala* : *falou* = *falava* : *tinha falado* and *fala* : *falava* = *falou* : *tinha falado*. Accordingly, only two binary features are required to account for these oppositions. One of them is [$\pm$ finished], which generates the relation *fala* : *falou* = *falava* : *tinha falado*. The other is [$\pm$ VP $\in$ past], which generates the relation *fala* : *falava* = *falou* : *tinha falado* and which indicates whether the speaker's temporal viewpoint is located within the NON-PAST interval or within the PAST interval. Thus, in Portuguese, the division between the NON-PAST and PAST intervals based on the speech time and the placement of the temporal viewpoint are inseparable, jointly constituting a single feature.

2.1.1 Present Indicative vs. Preterite Indicative — [$\pm$ finished]

These two forms are considered to constitute a semantic opposition in their original meanings expressed as [$-$ finished] vs [$+$ finished]. Let us now examine some examples:

(15) *O Rui está doente.* "Rui is ill."

(16) *A Maria fuma.* "Maria smokes."

(17) *Hoje / Amanhã ele vem aqui.* "He is coming here today / tomorrow."

(15) refers to the state *o Rui estar doente* as holding at the speech time. (16), which is often called the *habitual present*, is a use that expresses a habitual situation at the speech time. In other words, it refers to the event *a Maria fumar*, which occurs repeatedly and habitually within an indefinite time interval that includes the speech time but whose boundaries are unspecified. (17) refers, accompanied by a temporal adverbial phrase, to the single event *ele vir aqui*, which is set to occur in the near future with certainty. Thus, the present indicative can receive different temporal interpretations depending on the *Aktionsart* of the predicate and its interaction with

accompanying adverbial phrases. What all three examples share is, however, the fact that the situation in question is viewed as not finished from the speech time. On the other hand, the following three examples, all involving the preterite indicative, indicate that the situation in question is viewed as already finished from the speech time:

(18) *O Rui esteve doente.* "Rui was ill."

(19) *A Maria fumou.* "Maria smoked."

(20) *Ontem / Hoje ele veio aqui.* "He came here yesterday / today."

In (18), the state *o Rui estar doente* is already over at the speech time. Likewise, in (19) and (20), the events *a Maria fumar* and *ele vir aqui* are viewed as already finished from the speech time.

Accordingly, as far as these six examples are concerned, the present indicative indicates that the situation in question is viewed as not finished from the speech time, whereas the preterite indicative indicates that the situation is viewed as already finished from that time. From this, we may say that the opposition between the present indicative and the perfect indicative in the six examples above is an opposition between [–finished] (i.e., "situation viewed as not finished") and [+finished] (i.e., "situation viewed as finished"), where the speech time serves as the temporal viewpoint for the description of the situation⁵. This distinction between whether the situation is confirmed as finished or not — that is, [±finished] — appears to be an empirically significant one for humans. This is because a situation that is confirmed as finished is, in other words, an entity upon which no further intervention is possible, whereas a situation that is not confirmed as finished remains an entity still open to potential intervention. For example, in (16) *A Maria fuma*, it is not necessary for Maria to be smoking at the speech time. In such cases, what the sentence expresses as a **present** habit is a repetition of individual **past** events of *a Maria fumar*. Even so, it is marked as [–finished] because, when viewed as a single overall situation or tendency, it is ongoing at the speech time, is potentially interruptible and is therefore open to intervention. In (19) *A Maria fumou*, by contrast, the past single event expressed by the sentence is no longer open to intervention at the speech time — for instance, by being interrupted. This is why it is marked as [+finished]. The linguistic encoding of the feature

⁵ It is also possible to employ the notation [±perfect] instead of [±finished]. However, as footnote 14 on p. 139 of Mateus et al. (2003) suggests, the term *perfect* is highly ambiguous. This is because it is used not only in the sense of "an event that is finished when viewed from a given point in time" — which is essentially synonymous with *finished* here — but also in the sense of "a resultant state, established by an event that finished prior to a given point in time, and continuing up to that point" (as in the English present perfect). For this reason, in order to avoid ambiguity, this study adopts the unambiguous [±finished].

[±finished], which — as we shall observe throughout the course of the analysis — governs the entire system of temporal anchoring in Portuguese, seems to reflect this simple principle. This primitive opposition is illustrated in (21) with the verb *falar* in its third-person singular forms:

(21)

[-finished]	[+finished]
fala	falou

At the same time, however, there are cases in which it is questionable whether the present indicative truly indicates [-finished]. One such case is seen in usage (22), commonly referred to as the *gnomic present* or *timeless present*⁶:

(22) *Nem tudo o que luz é ouro.* "Not everything that shines is gold."

In this case as well, if we assumed that the sentence expresses a state that holds at any point within a temporally unbounded interval extending indefinitely before and after the speech time, then the situation could be interpreted as not finished at the speech time, and thus as another instance of [-finished]. However, as the traditional labels *gnomic present* and *timeless present* suggest, (22) may also be interpreted as describing atemporally or supra-temporally — that is, unconstrained by any temporal perspective — the situation itself independently of whether it is finished or not. If no temporal perspective is present, then naturally the question of whether the situation is finished or not does not arise. Accordingly, the present indicative in this example should be interpreted as carrying no implication whatsoever concerning the finishedness of the situation — that is, as [0finished]. Alternatively, the following interpretation may also be possible: a clause such as *nem tudo o que luzir ser ouro*, which does not use a finite verb form as its predicate, is grammatically unacceptable; therefore, in order to render it well-formed, a finite verb form must be used; however, if one were to employ a finite verb form bearing overt markers of mood and temporal reference — such as *-va*, *-re*, *-ria*, *-e*, *-sse* or *-r* — some specific temporal reference would necessarily be introduced. Hence, the unmarked present indicative, which lacks such markers, is used simply because it avoids this implication. If the present indicative is selected for this reason, then it would not be appropriate to interpret it as implying [-finished], which presupposes a temporal relationship.

What becomes clear from (22) is that there are cases in which it is difficult to straightforwardly interpret the present indicative as [-finished]. Taking this into account, along with the fact that

⁶ On the *historical present* or *narrative present*, see Section 5.2.

in its typical usage the present indicative sets the speech time as its temporal viewpoint and stands in opposition to the preterite indicative, which is specified as [+finished], the present study adopts the interpretation that the present indicative is [–finished] in its primitive meaning, but that it takes the value [0finished] in contexts or situations — such as those involving the gnomic present — that require the absence of a temporal perspective.

On the other hand, there are also cases in which the preterite indicative cannot be straightforwardly interpreted as [+finished]:

(23) (*Nasci e* vivi *sempre em Coimbra*. "I (was born and) have always lived in Coimbra."

The sentence (23) presents the time span from birth up to the speech time as a single closed time interval and states with the help of the temporal adverb *sempre* that, within this interval, the state *eu viver em Coimbra* held at any temporal point. This does not, nevertheless, entail that the state *eu viver em Coimbra* is finished at the speech time. It is therefore a different usage from that found in (18) – (20). Still, in (23), it can be argued that a synecdoche⁷ arises in relation to the primitive meaning of the preterite indicative [+finished]. That is, this primitive value [+finished] — more analytically, [+delimited by a finishing point] — undergoes semantic extension, by means of synecdoche, to a more generic value of [+delimited], as in (24):

(24) synecdoche: [+finished] = [+delimited by a finishing point] → [+delimited]

Thus, from the standpoint of primitive meaning, interpreting the preterite indicative as [+finished] poses no significant issue.

Through the operation of this derived feature [+delimited], the interval stretching from the point denoted by *nasci* to the speech time becomes a closed temporal span, within which the situation *eu viver em Coimbra*, holding throughout the entire interval, loses the potential to admit new elements or change at either endpoint — that is, it acquires the property of *intervention impossibility*. As a result, the situation *Nasci e vivi sempre em Lisboa* comes to function as a stable component of the speaker's self-definition and is therefore likely to appear as a formulaic expression in self-introductions or biographical statements. In this sense, the use of the preterite indicative in this sentence must be analysed from a pragmatic point of view.

⁷ Synecdoche is a figurative expression in which meaning is extended based on a relationship of inclusion — that is, a relationship between a species and a class (cf. Tsuji 2002: 158). It involves indicating a subordinate concept using a form that indicates a superordinate concept, or indicating a superordinate concept using a form that indicates a subordinate concept. In this particular case [+delimited] is the superordinate concept and [+finished] — or [+delimited by a finishing point] — is a subordinate one.

This study adopts, therefore, a cognitive-semantic perspective that seeks to explain the polysemy exhibited by a given morpheme or word in terms of its "primitive meaning" or "basic sense" and the "derived meanings" or "extended sense" that arise from it via metaphor, synecdoche, metonymy and other types of figurative extension. The novelty of this paper may lie in the fact that such **figurative extensions are introduced** not at the lexical level, but **at the level of semantic features**. This, in turn, might offer one explanation for **how a densely packed system that is both closed and saturated can nevertheless exhibit semantic openness**. Building on this foundation, the analysis proceeds by reducing the semantic oppositions observed among finite verb forms to oppositions between primitive anchoring-aspect features regarded as their original meanings.

2.1.2 Present and Preterite vs. Imperfect and Pluperfect — $[\pm\text{finished}] \times [\pm\text{VPEpast}]$

As discussed in Section 2.1.1, the present and preterite indicatives express the primitive semantic opposition $[-\text{finished}]$ vs. $[\text{finished}]$, provided that they share a clear temporal viewpoint such as the speech time. The central claim of this section is that, once the speaker's temporal viewpoint is shifted to another point located within **the time interval extending backward but excluding the speech time** — namely, **the PAST** — the opposition between the present and preterite indicatives gives rise to a new opposition, that between the imperfect and pluperfect indicatives. Here, the PAST is understood as the interval opposed to **the NON-PAST**, the latter being defined as **the time interval that extends forward and includes the speech time**. We now turn to some illustrative examples:

(25) *Ela estava doente quando cheguei.* "She was ill when I arrived."

(26) *Ela tinha estado doente quando cheguei.* "She had been ill when I arrived."

(27) *Ele disse que ia à festa de carro.* "He said that he was going to the party by car."

(28) *Ele disse que tinha ido à festa de carro.* "He said that he had gone to the party by car."

In (25), the use of the imperfect indicative *estava* indicates that the state *ela estar doente* was ongoing and had not yet finished at a certain point located in the PAST — namely, *quando cheguei* "when I arrived". By contrast, in (26), the use of the pluperfect indicative *tinha estado* indicates that the state *ela estar doente* had already finished by the past point *quando cheguei*. In (27), the use of the imperfect indicative *ia* indicates that the event *ele ir à festa de carro* was located in the future relative to the time of *ele disse*, which itself corresponds to a certain point

in the past. Thus, the event *ele ir à festa de carro* was still unfinished at the time of *ele disse*. In (28), by contrast, the use of the pluperfect indicative *tinha ido* shows that the event *ele ir à festa* had already finished relative to the past point *ele disse*.

In sum, the imperfect indicative in (25) and (27) indicates that, from a certain point in the PAST, the situation was viewed as not yet finished, whereas the pluperfect indicative in (26) and (28) denotes that the situation had already finished when viewed from such a past point. Accordingly, the aspectual opposition between the imperfect and pluperfect indicatives can be represented as the opposition between [–finished] and [+finished], where the temporal viewpoint for describing the situation is set at a certain point in the PAST. In what follows, this shared past temporal perspective will be denoted as [+VP∈past]. The element [past] in this feature refers to the PAST defined above — namely, the time interval extending backward without including the speech time. Its negative value [–VP∈past] is equivalent to [+VP∈non-past].

According to Mateus et al. (2003: 139–140), sentences such as *?O Rui lia o livro* "Rui was reading the book" and *?O Rui lera/tinha lido o livro* "Rui had read the book" are considered problematic in terms of acceptability. They are judged acceptable only when accompanied by a subordinate clause situating the event in a past temporal context, as in *O Rui lia o livro quando a Maria chegou* "Rui was reading the book when Maria arrived" or *O Rui lera/tinha lido o livro quando os amigos telefonaram* "Rui had read the book when his friends called". In other words, past time is generally indicated only through contexts and sentences with the imperfect or pluperfect indicative that lack such context are perceived by native speakers as problematic in terms of acceptability. This observation further supports the validity of interpreting the imperfect and pluperfect indicatives as bearing the feature [+VP∈past].

On the other hand, while the present and preterite indicatives were specified only in terms of [±finished] in Section 2.1, they normally stand in opposition to the imperfect and pluperfect indicatives, specified as [+VP∈past], since in their primitive sense they describe a situation from a temporal viewpoint located within the NON-PAST interval — whose default value is the speech time — and are thus specified as [–VP∈past]:

(29) *A Teresa é médica.* "Teresa is a doctor."

(30) *A Teresa foi médica.* "Teresa was a doctor."

(31) *A Teresa era médica quando a encontrei.* "Teresa was a doctor when I met her."

(32) *A Teresa tinha sido médica quando a encontrei.* "Teresa had been a doctor when I met her."

In both (29) and (31), the state *a Teresa ser médica* is viewed as ongoing from the speaker's temporal viewpoint and is thus marked as [–finished]. In (29), however, that viewpoint is located at a point within the NON-PAST interval — more specifically, at the speech time — and the finite verb *é* is therefore marked as [–VP∈past]. In contrast, in (31), the temporal viewpoint is located at a point within the PAST interval, namely the moment at which the event *eu a encontrar* occurred; the finite verb *era* is hence marked as [+VP∈past]. In both (30) and (32), on the other hand, the same state is viewed as no longer holding from the temporal viewpoint and is therefore marked as [+finished]. In (30), this viewpoint is placed at the speech time and the finite verb *foi* is accordingly marked as [–VP∈past]. In (32), by contrast, the temporal viewpoint is located at the past point *quando a encontrei* and the finite verb *tinha sido* is consequently marked as [+VP∈past]. On the basis of this observation, the present and preterite indicative forms are to be specified not only for [±finished] but also for [–VP∈past], as shown in (33) — here using the relevant forms of the verb *amar* "to love" for subsequent comparison with Classical Latin:

(33)

	[–finished]	[+finished]
[–VP∈past]	ama	amou
[+VP∈past]	amava	tinha amado

To our knowledge, it has not been explicitly pointed out that the entire assertive indicative paradigm of Portuguese — *ama, amou, amava, tinha amado* — can be exhaustively generated by the two binary features [±VP∈past] × [±finished]. This minimal 2×2 grid constitutes the backbone of the entire system and explains why the present perfect indicative — *tem amado* — could never be incorporated into this core paradigm (cf. Sections 0 and 5.1).

Historically speaking, until the emergence of compound tenses, the Portuguese indicative fact-marking paradigm was exhaustively saturated by the four simple forms *ama, amou, amava, amara* — precisely the 2×2 grid generated by [±past] × [±finished] or [±VP∈past] × [±finished], as shown in (34). As will be discussed in Section 5.7, it remains indeterminate at the present stage of our research whether, at that time, the Classical Latin tense feature [±past] had already shifted to the perspective feature [±VP∈past], characteristic of Modern Portuguese. For this reason, the notation [±(VP∈)past] is adopted in (34) to denote either [±past] or [±VP∈past]. It is clear from the difference in temporal-aspectual forms selection in counterfactual clauses that Classical Latin employed [±past], whereas Modern Portuguese employs [±VP∈past], as discussed in Sections 5.5 and 5.7.

(34)

	[-finished]	[+finished]
[-(VP€)past]	ama	amou
[(VP€)past]	amava	amara

However, due to the sound changes that brought the final nasal diphthong [-ẽũ], the third-person plural form *amaran/-am* of the pluperfect indicative became homophonous with *amaron/-om* — spelled *amaram* in modern orthography — of the preterite indicative. To avoid this homophony, the former is assumed to have been replaced by the periphrastic form *haviam amado*. As this replacement extended to the entire paradigm of the pluperfect indicative, it is thought that *amara* was ultimately replaced by *tinha amado* via *havia amado*. The replacement of *amara* by *havia/tinha amado* should thus not be understood as the result of a systemic reorganisation of the semantic relations among terms within the system.

Remarkably, **the core assertive indicative system in modern Portuguese defined by [±VP€past] × [±finished] has remained essentially unchanged since the time of Classical Latin, whose system was constituted by [±past] × [±finished], insofar as both formally encode the aspectual feature [±finished]**. This is exemplified in (35) by the Latin *præsens*, *perfectum*, *imperfectum* and *plusquamperfectum* in the third person singular of the verb *amāre* "to love":

(35)

	[-finished]	[+finished]
[-past]	amō	amāvit
[+past]	amābat	amāverat

Thus, **the Portuguese *Pretérito Perfeito Simples* remains a direct heir of the Latin *Perfectum* in its primary sense of finishedness without reference to a result state**: what appears today as a "past tense" is, in fact, nothing other than the finishedness-encoding form preserved since Latin times. The fact that the preterite indicative in Portuguese has functioned not as a *preterite* but as a *perfect* accounts for the semantic reanalysis of the *Pretérito Perfeito Composto*, as discussed in Sections 0 and 5.1.

Returning to the main point, the above feature specification — the preterite as [-VP€past, +finished] and the imperfect as [+VP€past, -finished] — directly accounts for the acceptability and naturalness of (36), the ungrammaticality of (37) and the markedness of (38):

- (36) *A Teresa era médica quando a encontrei, e ainda o é / mas já não o é.* "Teresa was a doctor when I met her and she still is / but she no longer is."

(37) **A Teresa foi médica, e ainda o é.* "Teresa was a doctor (but no longer is) and she still is."

(38) ?*A Teresa foi médica, mas já não é.* "Teresa was a doctor (but no longer is), but she no longer is."

In (36), the imperfect *era*, marked as [+VP_{past}, –finished], merely indicates that the state *Teresa ser médica* held at a certain point in the PAST — namely, the point referred to by *quando a encontrei* — without making any reference to what happened to that state afterward. Thus, it remains possible that the state ended before the speech time, but it is equally possible that it continued up to it. For this reason, *e ainda o é* "and she still is" and *mas já não o é* "but she no longer is" are both compatible continuations. In contrast, in (37) and (38), the preterite indicative *foi*, marked as [–VP_{past}, +finished], means that the state *Teresa ser médica* is confirmed as already finished at the speech time. Therefore, *e ainda o é* "and she still is" cannot be conjoined without a logical contradiction. On the other hand, the continuation *mas já não o é* "but she no longer is" is logically equivalent to *foi médica* "she used to be a doctor" and is thus semantically redundant — a redundancy that accounts for the unnaturalness of (38).

Earlier, we stated that the distinction [±finished] — that is, whether the finishedness of a situation is confirmed or not — is essentially significant for humans, since a situation whose finishedness is confirmed is one upon which no further intervention is possible, whereas a situation whose finishedness is not confirmed remains one open to intervention. Likewise, the distinction between the temporal intervals of the NON-PAST and the PAST on which the temporal viewpoint for describing the situation is located — i.e., [±VP_{past}] — is also fundamentally important, for the PAST represents a world in which one can no longer live and upon which no further intervention is possible, while the NON-PAST represents a world in which one is currently living or may still live and which is thus open to potential intervention. In other words, the opposition between the preterite indicative and the present indicative may be understood as the linguistic encoding of this fundamental cognitive distinction between situations upon which intervention is no longer possible and those upon which it remains possible, as seen from a world in which such intervention is still viable.

Returning to the the imperfect indicative and the pluperfect indicative, there are also cases in which they do **not** express [+VP_{past}]. One such case is their use in the consequent clause of a counterfactual conditional, as in (39):

(39) *Se eu tivesse mais dinheiro, comprava / tinha comprado isto.*⁸

"If I had more money, I would buy this." / "If I had had money, I would have bought this."

For the speaker, both "the past world" and "the non-real world" are conceived as alternative worlds opposed to "the non-past world" and "the real world", to which the speaker and hearer actually belong and which are open to human intervention. If the latter two are referred to more generically as "this world", then the former two constitute "another world", which can be accessed only through memory or imagination and which is not open to human intervention. Thus, these two alternative worlds — the PAST and the NON-REAL — jointly constitute a higher-order category: *another world* or *intervention impossibility*, which is encoded morphologically by the imperfect and the pluperfect. In other words, the perspective feature [+VP€past] undergoes metaphorical extension⁹ through the schema¹⁰ of *another world* or *intervention impossibility*, which subsumes both the PAST and the NON-REAL, to yield [+VP€non-real]. This figurative extension makes it possible to describe situations located in a non-real world from a cognitive viewpoint situated within that very world. Considered in this light, positing [+VP€past] as the primitive value presents no theoretical difficulty.

On the other hand, although the present indicative and the preterite indicative may be considered to stand in opposition to these forms in terms of a different temporal perspective — that is, as [–VP€past] — there also exist cases in which it is difficult to interpret either of these two forms as bearing the feature [–VP€past]. In the typical uses of the present indicative exemplified in (15) – (17) in section 2-1, the temporal viewpoint is located at the speech time, which falls within the non-past interval.

However, in the use commonly referred to as the gnomic present, exemplified in (22), the possibility arises that no temporal perspective is present at all. Furthermore, in uses of the

⁸ In the Spanish equivalents *Si yo tuviera más dinero, compraría esto* and *Si yo hubiera tenido más dinero, habría comprado esto*, unlike in the Portuguese example (31), the tense of the subordinate clause varies depending on whether the counterfactual situation concerns the past or the non-past. As will be discussed later, this difference stems from a conceptual distinction between a perspective-and-aspect-driven temporal-anchoring system (Portuguese) and a tense-driven temporal-anchoring system (Spanish, French, and Italian).

⁹ Metaphor is the use of a linguistic form that conventionally refers to one entity or concept to refer to another, based on some perceived similarity between the two. For example, in the phrase *the flower of the workplace*, the word *flower* refers not to a botanical organ, but to a person, mediated by the shared feature of *visually striking beauty* (cf. Tsuji 2002:16–17).

¹⁰ Schema is a commonality shared between two similar entities or concepts (cf. Tsuji 2002: 124).

present indicative found in dictionary definitions — such as *brilhante: que brilha* "brilliant: that shines" — the temporal perspective, as with the gnomic present, either does not exist or, if it does, the viewpoint may be located at any point on the temporal axis. It is therefore not appropriate to interpret these uses as bearing the feature $[-VP \in \text{past}]$. The same applies to the preterite indicative. For instance, in definitions found in dictionaries — such as *perdido: que se perdeu* "lost: that was lost" — where the preterite is used, the temporal viewpoint may be located at any point on the temporal axis and therefore can fall within either the PAST or the NON-PAST interval. What this use of the preterite does, then, is indicate the finishedness of the situation as viewed from that arbitrary temporal viewpoint.

On the basis of the above, it may be concluded that while the present indicative and the preterite indicative are, in their primitive senses, specified as $[-VP \in \text{past}]$, it is appropriate to interpret them as bearing the value $[0VP \in \text{past}]$ in special contexts or situations that require the arbitrary selection, ubiquity, or absence of a temporal perspective. As for counterfactual constructions, even when a metaphorical extension occurs — from $[+VP \in \text{past}]$ to $[+VP \in \text{non-real}]$ — via the above introduced schema of *another world* or *intervention impossibility*, **the present and the present perfect still bear the negative value $[-VP \in \text{non-real}]$** . In other words, they indicate that **the viewpoint is located within the real world** and are therefore **incompatible with the expression of counterfactuality**.

Thus far, the analysis has proceeded without positing any tense features such as $[\pm \text{past}]$, on the assumption that in this language, **the temporal localisation of a situation is derivatively computed from the feature value combination of $[\pm VP \in \text{past}]$ and $[\pm \text{finished}]$** . For instance, in the case of the preterite indicative, which bears the features $[-VP \in \text{past}]$ and $[+ \text{finished}]$ here, the situation as a whole is necessarily located in the past, i.e., prior to the speech time. On the other hand, in the case of the imperfect indicative, which bears the features $[+VP \in \text{past}]$ and $[- \text{finished}]$, at least the development of the situation up to the past point at which the temporal viewpoint is set is located in the past. It seems that this is a major characteristic of temporal localisation in Portuguese and, as we shall see later, it constitutes a significant point of divergence from other Romance languages such as Spanish and French. Accordingly, in what follows, the analysis will likewise proceed without positing any independent tense features.

2.2 Assumption-Marking Tense Forms: Non-future Forms vs. Future Forms — $[\pm\text{finished}] \times [\pm\text{VP}\in\text{past}] \times [\pm\text{assumptive}]$

In Section 2.1.2, we saw that the four basic tense forms used to present a situation as factual are exhaustively generated by the two binary features $[\pm\text{finished}] \times [\pm\text{VP}\in\text{past}]$. In this section, we shall show that by superimposing the modal feature $[\pm\text{assumptive}]$ onto this core system, the future tense forms are recursively derived.

The present, preterite, imperfect and pluperfect indicative forms, as seen in (40) – (43), all indicate that the situation described is taken by the speaker to be factual:

(40) *Ele agora está em Lisboa.* "He is now in Lisbon."

(41) *A Ana ainda não chegou. Perdeu o autocarro.* "Ana hasn't arrived yet. She missed the bus."

(42) *O Nuno tinha vinte anos quando o encontrei.* "Nuno was twenty years old when I met him."

(43) *Quando elas chegaram ao porto, o barco já tinha saído.* "When they arrived at the harbour, the boat had already left."

By contrast, the future, future perfect, past future and past future perfect indicative forms — as illustrated in (44) – (47) — typically express, particularly in formal written registers, a doxastic possibility on the part of the speaker (i.e., an assumption), while maintaining the same temporal reference as their factual counterparts (40) – (43):

(44) *Ele agora estará em Lisboa.* "I believe he is in Lisbon now."

(45) *A Ana ainda não chegou. Terá perdido o autocarro.* "Ana hasn't arrived yet. I believe she missed the bus."

(46) *O Nuno teria vinte anos quando o encontrei.* "I believe Nuno was twenty years old when I met him."

(47) *Quando elas chegaram ao porto, o barco já teria saído.* "When they arrived at the harbour, I believe the boat had already left."

That is, while the present, preterite, imperfect and pluperfect indicative forms — when used without modal expressions — typically present the situation referred to by the clause as an actual event or state, the future, future perfect, past future and past future perfect indicative forms present the same situation, with the same temporal localisation, as being more likely the case, based on the speaker's estimation.

Assumption is a type of modality whereby the situation referred to by the sentence is not asserted as an actual fact, but rather as something that is possibly the case. However, the kind of possibility expressed by the future tense is not the same as that expressed by the modal verb *poder* in (48):

(48) *Ele agora pode estar em Lisboa.* "He may be in Lisbon now."

This is because the sentence in (48) can be paraphrased as *Ele agora pode estar ou pode não estar em Lisboa* "He may or may not be in Lisbon now", whereas (44) *Ele agora estará em Lisboa* "I believe he is in Lisbon now" cannot be paraphrased as **Ele agora estará ou não estará em Lisboa* "*He will be or will not be in Lisbon now". This is due to the fact that, in (48), the speaker treats the possibility of the occurrence of *ele agora estar em Lisboa* and that of *ele agora não estar em Lisboa* as equally plausible, taking no stance on either. In contrast, in (44), the speaker places some degree of belief in the possibility of the occurrence of *ele agora estar em Lisboa*, while withholding belief from that of *ele agora não estar em Lisboa*. This asymmetry is further confirmed by the fact that (44) can be paraphrased as *Creio que ele agora está em Lisboa* "I believe he is in Lisbon now". In this way, **assumption** refers to **the speaker's belief, to some degree, in the possibility of the occurrence of a given situation**. In the present study, this type of possibility — normally called doxastic possibility —, as expressed by the future tense forms in the indicative mood, is posited as their primitive meaning and is represented as [+assumptive]. As we have seen, the future, future perfect, past future and past future perfect indicative each establish an identical temporal anchoring to that of the present, preterite, imperfect and pluperfect indicative, respectively. Therefore, apart from the opposition in [±assumptive], they share the same feature values for [VP∈past] and [finished]. The combination of the three primitive features established thus far, with each taking either a positive or negative value, both logically and actually yields eight finite verb forms. These are exemplified in (49), using the verb *falar* in its third person singular forms:

(49)

	[−finished]	[+finished]	[−finished]	[+finished]
[−VP∈past]	fala	falou	falará	terá falado
[+VP∈past]	falava	tinha falado	falaria	teria falado
	[−assumptive]		[+assumptive]	

The feature [+assumptive], proposed above, also accounts for the use of the future indicative to express deontic modality, as exemplified in (50):

(50) *Não matarás.* "You shall not kill."

In such cases, a metonymic extension¹¹ — a type of figurative process — appears to occur: from "I **believe** you will not kill" (a belief serving as cause or justification) to "(therefore) you **must** not kill" (a command as expected effect or consequence). Here, a form that originally expresses a cause comes to stand for its result — a mechanism commonly attested in world languages.

On the other hand, while the uses of the future tense forms as in (44) – (47) and (50) are generally referred to as *modal*, the following uses in (51) and (52) are commonly classified as *temporal*:

(51) *A Ana casará daqui a dois meses.* "Ana will get married in two months."

(52) *O Jorge disse-nos que voltaria daí a uma semana.* "Jorge told us that he would return in a week."

However, when examples (51) and (52) are compared with their non-future indicative counterparts (53) and (54), it becomes clear that the temporal anchoring of the situation is, in fact, exactly the same:

(53) *A Ana casa daqui a dois meses.* "Ana gets/is getting married in two months."

(54) *O Jorge disse-nos que voltava daí a uma semana.* "Jorge told us that he was returning in a week."

In both (51) and (53), *casará* and *casa* indicate a situation located in the future relative to the speech time — *daqui a dois meses* "in two months". Similarly, in both (52) and (54), *voltaria* and *voltava* refer to a situation located in the future relative to the PAST point at which the event *o Jorge dizer-nos* occurred — *daí a uma semana* "in a week". In Portuguese, therefore, the future-oriented reference — i.e., temporal localisation of a situation in the future — is not a function exclusive to the future tenses. As for the specification in (49), both the present indicative and the future indicative share the values [–VP∈past, –finished], opposed solely in [±assumptive]. Their common primitive function is thus, nowadays, to describe **a situation whose finishedness cannot be confirmed from a temporal viewpoint located at some point within the non-past interval — typically, its default value, the speech time.** Since a future

¹¹ Metonymy refers to the use of a form that refers to one entity or concept in order to refer to another, based on their contiguity in the external world or, more broadly, on their cognitive or conceptual association. For example, in the expression "to read Eça de Queirós", it is not the author himself but rather his works that are referred to by the name, based on the conceptual relationship between "author" and "work" (cf. Tsujii 2002, pp. 35–36).

situation, viewed from the speech time, has not even begun, its finishedness cannot be confirmed either. For this reason, such situations can be described not only by the future indicative but also by the present indicative. In modern Portuguese, the widespread use of the present indicative to describe future situations — insofar as it satisfies the condition of [–assumptive]¹² — is due precisely to its feature values [–VP∈past, –finished]. Crucially, this usage does not represent a semantic extension of the present tense, but rather constitutes its primitive function. In parallel, both the imperfect and the past future indicative share the values [+VP∈past, –finished], opposed only in [±assumptive]. Their common primitive function is therefore to describe **a situation whose finishedness cannot be confirmed from a temporal viewpoint located at some point within the past interval**. Since a past future situation, viewed from a given point in the past, has not even begun at that point, its finishedness cannot be confirmed either. This is why such situations can be described not only by the past future indicative but also by the imperfect indicative. The widespread use of the imperfect indicative to describe situations in the past future — as far as it meets the condition of [–assumptive]¹³ — is due precisely to its feature values [+VP∈past, –finished]. This usage does not represent a semantic extension of the past tense, but constitutes instead its primitive function.

What is interesting in contrast is the use of the past future indicative in (55):

- (55) *Ele começou como líder estudantil e, mais tarde, seria / *era primeiro ministro.* "He started out as a student leader and would later become prime minister."

In this case, unlike in (54), the use of the imperfect indicative is not possible. The fundamental difference between the past future situation expressed by *seria* in (55) and that expressed by *voltava* in the imperfect indicative in (54) lies in the fact that *seria* in (55) refers to a past future situation as a **fact**, whereas with *voltava* in (54), it remains unclear whether the past future situation is being presented as an actual fact. This is because (55) indicates that the state *ele ser*

¹² When the feature [–assumptive] cannot be maintained in the expression of a future situation, in place of the present indicative, the periphrastic construction *ir* in the present indicative + infinitive is generally used in contemporary spoken Portuguese. The essential function of this periphrastic form is to situate the event in future time. Consequently, it cannot situate the event in the present. By contrast, the future indicative, which is marked today as [–VP∈past, –finished, +assumptive], has as its essential function the assumptive indication of non-finished situations located in the present or the future. Since its function is not essentially temporal, it is not generally selected as the primary option in such contexts.

¹³ When the feature [–assumptive] cannot be maintained in the expression of a past future situation, in place of the imperfect indicative, the periphrastic construction *ir* in the imperfect indicative + infinitive is generally used in contemporary spoken Portuguese. In this register, the past future indicative is not generally selected as the primary option in such contexts, for the same reason as the future indicative.

ministro actually came to hold later as a fact, whereas in (54), the event *o Jorge voltar daí a uma semana* may be no more than a verbal commitment on Jorge's part and this does not pose any problem. Thus, the past future indicative in (55) serves not only to locate the situation in the future relative to a past point expressed by the first clause *ele começou como líder estudantil*, but also to explicitly mark its factuality. Both the past future indicative and the imperfect indicative share the modal feature [–non-assertive], whose original meaning is the indication of factuality — as will be explained in Section 2.4, the tense forms of the indicative mood contrast with those of the subjunctive mood in terms of the feature [$\pm$ non-assertive]. The factuality of *ele ser ministro* in (55) can be indicated by the feature [–non-assertive] and what remains to be accounted for is the indication of futurity. This is achieved through a metonymic semantic extension whereby the feature [+assumptive], originally modal, is extended to express the temporal feature [+posterior] in the past future indicative. A situation located in the future relative to a given point in time cannot, by definition, be known as a fact at that point; it must necessarily be the object not of knowledge, but of inference — such as assumption. In other words, [+posterior] and [+assumptive] stand in a conditional-consequential relation: if a situation is located in the future (the *condition*), then it must necessarily be treated as an object of assumption (the *consequence*). It is on the basis of this relation that the feature [+assumptive] is considered to have undergone a metonymic semantic extension to [+posterior], as in (56):

(56) metonymy: modal [+assumptive] (*consequence*) → temporal [+posterior] (*condition*)

By contrast, the imperfect indicative, specified as [–assumptive], leaves no room for such a metonymic semantic extension. It therefore cannot indicate futurity in contexts such as (55). From the above discussion, it can be concluded that even in examples like (55), the original meaning of the past future indicative may be posited as [+assumptive].

By contrast, while the present, perfect, imperfect and pluperfect indicative forms in (40) – (43) present situations as facts, there are cases in which interpreting them as [–assumptive] becomes problematic. For example, in the sentence *Se calhar ele é brasileiro* ("He's probably Brazilian"), the function of [+assumptive] is carried by *se calhar*. However, if the present indicative *é* were [–assumptive], the sentence would imply the contradictory proposition that *ele ser brasileiro* is both a fact and an assumption. On the other hand, if *é* is interpreted as [0assumptive], the sentence would simply mean that *ele ser brasileiro* is an assumption, which is not problematic. It is therefore reasonable to interpret the present, perfect, imperfect and pluperfect indicative forms as being originally [–assumptive], but capable of taking a [0 assumptive] value when accompanied by modal expressions such as adverbial phrases.

2.3 Non-Assertive Tense Forms: Indicative vs. Subjunctive — $[\pm\text{finished}] \times [\pm\text{VP}\in\text{past}] \times [\pm\text{assumptive}] \times [\pm\text{non-assertive}]$

This section examines the semantic opposition between the indicative and subjunctive moods. We shall see that, by superimposing the mood feature $[\pm\text{non-assertive}]$ onto the full indicative system — which, as shown in Section 2.2, is generated by $[\pm\text{finished}] \times [\pm\text{VP}\in\text{past}] \times [\pm\text{assumptive}]$ — the subjunctive tense forms are recursively generated.

(57) *Ela hoje veio / *tenha vindo aqui.* "She came here today."

(58) *Espero que ela hoje *veio / tenha vindo aqui.* "I hope she has come here today."

In (57), the preterite indicative *veio* is used and the speaker is, by means of this verbal form, asserting the event *ela hoje ter vindo aqui* as a fact. In other words, the speaker makes a truth-value judgment as to whether the situation described by the sentence is factual and judges it to be true. Consequently, if it turns out that she has not actually come, the speaker would be regarded as having made a false statement. On the other hand, in (58), the occurrence of *ela hoje ter vindo aqui* represents merely the speaker's wish and the speaker is not in a position to guarantee whether it is actually the case. That is, the speaker must suspend a truth-value judgment and it is precisely for this reason that the preterite subjunctive *tenha vindo* is used. (57) and (58) are examples in which the indicative and the subjunctive cannot be interchanged. However, there are also cases in which both moods can be used, though with a clear difference in meaning:

(59) *A Maria quer casar com um espanhol que sabe falar português.* "Maria wants to marry a Spanish man, who can speak Portuguese."

(60) *A Maria quer casar com um espanhol que saiba falar português.* "Maria wants to marry a Spanish man who can speak Portuguese."

In (59), *um espanhol que sabe falar português* refers to an actually existing individual. Since he exists, his ability — *saber falar português* — is taken as a fact and the present indicative *sabe* is used in order to assert that factuality. In contrast, *um espanhol que saiba falar português* in (60) refers to someone whom Maria merely imagines as an ideal partner and the speaker is unable to assert whether such a person actually exists. If the indicative *sabe* were used in this case, it would present such a person as actually existing, as in (59). The present subjunctive *saiba* is thus used in (60).

As such, the indicative is the verbal form that explicitly expresses the speaker's judgment as to

whether the situation described by the sentence is a fact. In other words, it is a verbal form that indicates a truth-value judgment, or, alternatively, the verbal form of assertion. In contrast, the subjunctive is the verbal form by which the speaker suspends judgment regarding the factual status of the situation to be described. In other words, it is a verbal form that suspends truth-value judgment, or, alternatively, the verbal form of non-assertion. There are two types of motivation for the suspension of a truth-value judgment. One is when such a judgment is not possible, as in (58) and (60). The other is when a truth-value judgment is not required:

(61) *Sinto que o Gonçalo esteja doente.* "I feel sorry that Gonçalo is ill."

(61) expresses that *eu* is emotionally reacting to the situation *ele estar doente*. Since situations that trigger human emotional reactions are generally taken to be factual, there is no need to distinguish whether the situation is real or hypothetical. Its factuality is *presupposed* (cf. Perini 2001: 184–186). Therefore, it is not necessary to use the indicative, which would assert the factuality of the situation; instead, the subjunctive *esteja* is used. On the other hand, when the same verb *sentir* is used not as a verb of emotion but as a verb of perception, as in (62), the indicative appears in the subordinate clause:

(62) *Sinto que o Gonçalo está doente.* "I feel that Gonçalo is ill."

Thus, the opposition between the subjunctive and the indicative can generally be described as an opposition between truth-value judgment — assertion — and the suspension of truth-value judgment — non-assertion. Accordingly, we shall represent this opposition using the features [–non-assertive] and [+non-assertive]. However, there are cases where the indicative cannot easily be interpreted as [–non-assertive]:

(63) *Talvez aquele seja chinês.* "That person may be a Chinese."

(64) *Aquele é talvez chinês.* "That person may be a Chinese."

In these two examples, *talvez* expresses a modality of speculation, [+speculative], similar to *poder* in (48). In (63), the present subjunctive *seja*, marked as [+non-assertive], is semantically compatible with *talvez*'s [+speculative] modality and the sentence as a whole conveys the meaning "it is speculated that he is Chinese". In contrast, if in (64) the indicative present *é* is interpreted as [–non-assertive], the sentence would result in the contradictory proposition that *aquele ser chinês* is both asserted as a fact and simultaneously presented as a speculation. However, if *é* is instead assigned the neutral value [0 non-assertive], (64) simply expresses "it is speculated that he is a Chinese" and no contradiction arises. Therefore, while the indicative is

primitively [−non-assertive], it may contextually take the value [0 non-assertive], depending on the presence of other modal elements. The combination of the four primitive features established so far, each taking either a positive or a negative value, logically yields 16 finite verb forms. Of these, two highly marked forms specified as [+non-assertive, +VP∈past, ±finished, +assumptive] are excluded, resulting in a total of 14 actual forms. These are exemplified in (65), using the verb *falar* in its third-person singular forms. The most marked area in the system are shaded and the areas not filled with actual forms are indicated with strikethrough:

(65)

	[−finished]	[+finished]	[−finished]	[+finished]	
[−VP∈past]	fala	falou	falará	terá falado	[−non-assertive]
[+VP∈past]	falava	tinha falado	falaria	teria falado	
[−VP∈past]	fale	tenha falado	falar	tiver falado	[+non-assertive]
[+VP∈past]	falasse	tivesse falado	————	————	
	[−assumptive]		[+assumptive]		

Finally, let us turn to the future subjunctive *falar* and the future perfect subjunctive *tiver falado*. Unlike the corresponding future forms in the indicative mood — *falará*, *terá falado*, *falaria*, *teria falado* — these two forms are characterised, even in ordinary spoken language, by what Teyssier (1985: 219) describes as "le sens de ce temps est celui de future." Accordingly, in these two forms, the primitive anchoring-aspect feature [+assumptive] common to all future forms undergoes — as shown in (66) = (56) — a metonymic extension via a condition-consequence relation, yielding the derived feature [+posterior] (cf. Section 2.2), which clearly establishes a reference time in the future. It is this derived feature that is primarily operative in these forms:

(66) metonymy: modal [+assumptive] (*consequence*) → temporal [+posterior] (*condition*)

The reason is simple: both [+non-assertive] — the subjunctive mood itself — and [+assumptive] — the feature common to all the future forms — are functionally equivalent, in that they both serve to avoid assertion. If [+assumptive] were retained unchanged, semantic redundancy would arise. The derived feature [+posterior], by contrast, does not overlap functionally with [+non-assertive]. As Araújo (2003: 141) observes, "À la valeur temporelle du futur s'ajoute une valeur hypothétique donnée par le subjonctif", where *la valeur temporelle du futur* corresponds to [+posterior] and *une valeur hypothétique donnée par le subjonctif* corresponds to [+non-assertive]. As a result of this extension, the use of the future subjunctive alone comes to signal the conditional frame "if, at some point in the future, X happens" — without the need for

additional temporal or modal markers. This semantic efficiency likely contributed to the consolidation of the future and future perfect subjunctives within the Portuguese temporal anchoring system, as illustrated in (67) and (68):

(67) *Quando eles vierem, vamos jantar juntos.* "When they come, we will have dinner together." [+non-assertive, -VP∈past, +posterior, -finished]

(68) *Quando tiveres falado, podemos discutir isso.* "When you have spoken, we can discuss it." [+non-assertive, -VP∈past, +posterior, +finished]

In these examples, the temporal anchoring process is assumed to proceed as illustrated in (69):

(69) [-VP∈past] ⇒ [+posterior] (← [+assumptive]) ⇒ [±finished]

Note that, as a result of the metonymic shift (66), the modal feature [+assumptive], which is ordinarily employed at stage 3 of the process, is effectively reanalysed as the reference time feature [+posterior] and is thereby promoted to stage 2. Conversely, the aspectual feature [±finished], now demoted to stage 3, is employed to anchor the situation time relative to the future time marked as [+posterior].

2-5. Present vs. Present Perfect Indicative

The final contrast concerns the present indicative and the present perfect indicative:

(70) *Ela casou aos 18 anos e (desde então) tem levado uma vida cheia de dificuldades.* "She got married at 18 and (since then) has led a life full of hardships."

(71) *Ela casou aos 18 anos e (agora) leva uma vida cheia de dificuldades.* "She got married at 18 and (regardless of what may have occurred between then and now, at least at present) leads a life full of hardships."

In (70), which contains the present perfect indicative, the addition of *desde então* after the conjunction *e* yields a sentence that is virtually synonymous. That is, when the speech time serves as the temporal viewpoint, the present perfect indicative expresses a situation that has continued from a point in the past up to the speech time, or one that has been repeated an unspecified number of times during that interval. In (71), in contrast, which contains the present indicative, the insertion of *agora* after *e* also yields a sentence that is virtually synonymous. Unlike the present perfect indicative, however, the present indicative does not in itself imply any specific starting point for the situation. In other words, while the present perfect indicative

generally makes explicit that **the situation began at some point prior to the speech time**, the present indicative differs in that it **says nothing about when the situation began**. Accordingly, the contrast between the present indicative and the present perfect indicative in (70) and (71) can be represented as [0initiated] vs [✓initiated]¹⁴. In contrast, no such semantic opposition is observed among the other finite verb forms and therefore they can be considered to carry the neutral value [0initiated], that is, they make no statement regarding the inception point of the situation. (72) incorporates, within the structure of (65), the opposition between the present indicative, specified as [–non-assertive, –VP∈past, –finished, –assumptive, 0initiated, 0stative] and the present perfect indicative, specified as [–non-assertive, –VP∈past, –finished, –assumptive, ✓initiated, ✓stative] (on the *Aktionsart* feature [stative], see Section 5.1):

(72)	[–finished]		[+finished]	[–finished]	[+finished]	
[–VP∈past]	[0 initiated, 0 stative] fala		falou	falará	terá falado	[–non-assertive]
	[✓initiated, ✓stative] tem falado					
[+VP∈past]	falava		tinha falado	falaria	teria falado	[+non-assertive]
[–VP∈past]	fale		tenha falado	falar	tiver falado	
[+VP∈past]	falasse		tivesse falado	_____	_____	
	[–assumptive]			[+assumptive]		

Very interestingly, when the temporal viewpoint is not on the speech time, the present perfect indicative may be equivalent to the preterite indicative, as in (73), and is therefore specified as [–non-assertive, –VP∈past, +finished, –assumptive]:

(73) *Sempre que a Ana chega a casa da Maria, já o Rui a tem visitado / visitou.* "Whenever Ana arrives at Maria's house, Rui has already visited her" (from Mateus et al. 2003: 143).

In this case, the temporal viewpoint is not anchored to the speech time but to multiple points within the non-past interval, as indicated by the indefinite quantifier clause *Sempre que a Ana chega a casa da Maria*. Therefore, the present analysis using the feature [initiated] and [stative] is based solely on the most common usage in which only the speech time serves as the temporal viewpoint.

¹⁴ The symbols [+] and [–] represent an opposition between a given semantic value and its direct opposite, whereas [✓] and [0] indicate an opposition between the presence of a semantic value and its absence. If, in opposition to the present perfect indicative, the present indicative were specified as [–initiated, –state], then it would represent any situation as "an event without an initiation point". This privative notation parallels phonological underspecification, here applied to semantic features.

2.6 Some Problematic Cases to Analyse

In (74), Napoleon's invasion of Russia and his defeat are described using the preterite indicative:

(74) *Napoleão invadiu a Rússia em 1812 e sofreu uma derrota devastadora.* "Napoleon invaded Russia in 1812 and suffered a devastating defeat."

These situations are thus marked as $[-VP\in\text{past}, +\text{finished}]$, because they are finished when viewed from a viewpoint located at the speech time. In (75), however, the same past situations are narrated using the present indicative, specified as $[-VP\in\text{past}, -\text{finished}]$:

(75) *Napoleão invade a Rússia em 1812 e sofre uma derrota devastadora.* "Napoleon invades Russia in 1812 and suffers a devastating defeat."

This type of usage of the present indicative is commonly referred to as the *historical present* or *narrative present*. How can this be possible? The answer lies in the boundary between the NON-PAST and the PAST. For example, when the present indicative is used to express a *habitual present*, the boundary between the PAST and the NON-PAST can shift significantly backward from the speech time — reaching into what would ordinarily be considered the past in everyday terms, provided that the speech time remains included within the NON-PAST. In this usage of the present indicative, it may be assumed that the boundary is displaced as far to the left as necessary and that the speaker's viewpoint is placed at some point within this expanded NON-PAST, simultaneous with the narrated situations. From this viewpoint, the situations are described as if unfolding before the speaker's eyes — and accordingly, the verb that denotes the *situation-base* is marked as $[-\text{finished}]$ ¹⁵. From this, it may be concluded that $[-VP\in\text{past}]$ designates the location of the speaker's viewpoint within a temporally extended NON-PAST interval — one that is pragmatically stretchable leftwards (i.e., towards the past), so long as it includes the speech time. The primitive feature specification $[\pm VP\in\text{past}] \times [\pm \text{finished}]$, therefore, presents no problem even in cases like this. (*to be continued...*)

3. Summary — Feature Analysis of Portuguese Indicative and Subjunctive Forms

Based on the discussion in Section 2 and the feature chart in (72) from Section 2.5, the following generalisations can be drawn from the examples in (76) through (81):

¹⁵ The *situation-base* refers to the fundamental schematic structure of a situation — a structure that becomes an actual situation when participants are integrated into it. Conversely, a situation-base can be understood as what remains of a situation when its participants are abstracted away.

(76) The fundamental semantic oppositions among the various indicative and subjunctive forms in Portuguese can be captured using four primitive anchoring-aspect features: mood [non-assertive], perspective [VP $\in$ past], aspect [finished] and modality [assumptive], along with additional features — aspect [initiated] and *Aktionsart* [stative] — which applies only to the opposition between the present and present perfect indicatives.

(77) Although a feature may be specified negatively in its original sense, it can take the value [0] depending on contexts or/and situations. In addition, figurative extension may lead to the emergence of derived features from original ones (e.g. [VP $\in$ past] $\rightarrow$ [VP $\in$ non-real]; [assumptive] $\rightarrow$ [posterior], etc.).

(78) The combination of feature values with the lowest degree of semantic markedness in the system — namely, [–non-assertive, –VP $\in$ past, –finished, –assumptive] and [0initiated, 0stative] — is expressed by the present indicative. Its morphological structure serves as evidence of this semantic unmarkedness: the fact that all of its features are either negative or zero is reflected in the phonological absence of overt markers for mood and temporal anchoring.

(79) Combinations with the highest degree of semantic markedness — such as [+non-assertive, +VP $\in$ past, +finished, +assumptive] (a theoretical past future perfect subjunctive) and [+non-assertive, +VP $\in$ past, –finished, +assumptive] (a theoretical past future subjunctive) — do not exist. Given the cross-linguistic tendency for highly marked forms to be excluded from paradigms, the lack of such forms in Portuguese appears to be typologically well motivated.

(80) In the least semantically marked region of the system, namely [–non-assertive, –VP $\in$ past, –finished, –assumptive], the opposition between [0initiated] and [$\checkmark$ initiated] — which is encoded by that between the present and present perfect indicatives — can be observed. This opposition is unique to this domain and does not occur in the other regions of the system.

(81) The diagram in (72) can be reformulated as the following feature matrix:

	Non-assertive	VP€past	Initiated	Stative	Finished	Assumptive
Present Indicative						
Present Perfect. Indicative			✓	✓		
Preterite Indicative					+	
Imperfect Ind		+				
Pluperfect Indicative		+			+	
Future Indicative						+
Future Perfect Indicative					+	+
Past Future Indicative		+				+
Past Future Perfect Ind.		+			+	+
Present Subjunctive	+					
Preterite Subjunctive	+				+	
Imperfect Ind	+	+				
Pluperfect Subjunctive	+	+			+	
Future Subjunctive	+					+
Future Perfect Subjunctive	+				+	+
<i>Future Past Subjunctive</i>	+	+				+
<i>Future Past Perfect Subj.</i>	+	+			+	+

Note that, in order to indicate that forms specified as [–] may bear the value [0], the [–] value is left blank and that neither the past future subjunctive nor the past future perfect subjunctive actually exists; they are presented here merely as theoretical possibilities of feature combination.

4. Spanish and French — Tense-Driven Temporal Anchoring System

In what follows, Spanish will be taken as the representative of the tense-driven temporal system among the Romance languages and its system of temporal anchoring will be analysed concisely. Spanish has been chosen because, apart from phonetics and phonology, it is formally the closest to Portuguese among the major Romance languages and because this formal proximity conceals a number of differences in the way temporal anchoring is structured.

The tense forms commonly used in contemporary European Spanish are, in the indicative mood: present, present perfect, imperfect, preterite, pluperfect, future, future perfect, past future and past future perfect. The preterite anterior indicative is generally not used. In the subjunctive mood, the commonly used forms are: present, present perfect, imperfect and pluperfect. The future and future perfect subjunctive forms are generally no longer in use. The correspondences in terms of temporal reference between the tense forms of the indicative and those of the subjunctive are shown in (82), where forms with the same temporal reference are aligned on the same horizontal line (excluding counterfactual uses):

(82)

Indicative		Subjunctive
Present	Future	Present
Present perfect	Future perfect	Present perfect
Imperfect	Past future	Imperfect
Preterite		
Pluperfect	Past future perfect	Pluperfect

From (82), it must be concluded that the feature which distinguishes, in the indicative, between the non-future tenses as a whole and the future ones as a whole — namely, [$\pm$ assumptive], as we shall see later — is neutralised in the subjunctive. Hence, for example, the present perfect and future perfect indicatives as a whole correspond to the present perfect subjunctive, since **the subjunctive has no dedicated form for expressing futurity**. If the main clause verb *creo* in (83) and (84) is negated, the subordinate clause verbs *ha llegado* and *habrá llegado* change into the same form *haya llegado*, as seen in (83) – (86):

(83) *Creo que a estas horas ya ha llegado allí.* "I think by now he/she/you has already arrived there."

(84) *Creo que mañana a estas horas ya habrá llegado allí.* "I think by this time tomorrow he/she/you will already have arrived there."

(85) *No creo que a estas horas ya haya llegado allí.* "I don't think by now he/she/you has already arrived there."

(86) *No creo que mañana a estas horas ya haya llegado allí.* "I don't think that by this time tomorrow he/she/you will already have arrived there."

The above set of correspondences in (82) differs in a fundamental way from those found in Portuguese and this difference reflects a deeper structural contrast between the temporal-reference systems of the two languages. The key issue concerns **the correspondence between the imperfect and preterite indicatives and the imperfect subjunctive**. In Portuguese, the subjunctive forms that correspond to the preterite and imperfect indicatives are, respectively, the preterite and imperfect subjunctives. Hence, the opposition between [$-VP \in \text{past}$, $+\text{finished}$] and [$+VP \in \text{past}$, $-\text{finished}$] is formally encoded even within the subjunctive:

(87) *Ele foi médico.* "He was a doctor (it is specified that he is no longer one)." [$-VP \in \text{past}$, $+\text{finished}$]

(88) *Ele era médico quando o encontrei.* "He was a doctor when I met him (it is not specified whether he still is)." [$+VP \in \text{past}$, $-\text{finished}$]

(89) *Talvez ele tenha sido médico.* "Perhaps he was a doctor (it is specified that he is no longer one)." [–VP∈past, +finished]

(90) *Talvez ele fosse médico quando o encontrei.* "Perhaps he was a doctor when I met him (it is not specified whether he still is)." [+VP∈past, –finished]

The above opposition is encoded not only between (87) and (88) in the indicative, but also between (89) and (90) in the subjunctive.

In Spanish, by contrast, both the *preterite indicative* and the *imperfect indicative* correspond to a single form: the *imperfect subjunctive* (see Senction 5.5 on this difference between these two languages):

(91) *Pedro fue médico.* "Pedro was a doctor — it is specified that he is no longer one."

(92) *Pedro era médico cuando le encontré.* "Pedro was a doctor when I met him — it is not specified whether he still is."

(93) *Quizás Pedro fuera médico.* "Perhaps Pedro was a doctor — it is inferred that he is no longer one."

(94) *Quizás Pedro fuera médico cuando le encontré.* "Perhaps Pedro was a doctor when I met him — it is not specified whether he still is."

The aspectual opposition between the preterite *fue* in (91) and the imperfect *era* in (92) within the indicative is neutralised as *fuera* in (93) and (94) in the subjunctive and must instead be recovered from contextual or situational cues in this mood.

At the same time, the fact that both the preterite and the imperfect indicatives correspond to the same past subjunctive form suggests that — apart from the opposition between the indicative and the subjunctive and the aspectual opposition between the preterite and the imperfect, which we shall discuss later — the three forms *fue*, *era* and *fuera* are temporally equivalent in value.

(95) *Pedro había sido médico cuando le encontré.* "Pedro had been a doctor when I met him — he was no longer a doctor at the time I met him."

(96) *Quizás Pedro hubiera sido médico cuando le encontré.* "Perhaps Pedro had been a doctor when I met him — he was no longer a doctor at the time I met him."

In (92) and (94), the situation *Pedro ser médico* is presented — by the imperfect indicative and the imperfect subjunctive — as simultaneous with a past reference time denoted by *cundo le*

encontré, whereas the same situation is presented — by the pluperfect indicative and the pluperfect subjunctive — as anterior to that same past reference time in (95) and (96). This suggests that the imperfect indicative and the imperfect subjunctive are both characterised by the same combination of the absolute tense feature [+past] and the situation-time feature [–anterior] — that is, [+simultaneous] — in contrast to the pluperfect indicative and the pluperfect subjunctive, which are specified as [+past, +anterior]. Furthermore, the temporal relation between the preterite and the pluperfect indicatives is parallel to that between the imperfect and the pluperfect indicatives:

(97) *Pedro fue médico, pero antes había sido profesor*. "Pedro was a doctor, but before that he had been a teacher."

In (97), the situation *Pedro ser profesor* is presented by the pluperfect as anterior to a past reference time denoted by *Pedro fue médico* and is thus marked as [+past, +anterior], whereas the situation *Pedro ser médico* is — in terms of temporal logic — marked as [+past, –anterior] by the preterite.

Excluding counterfactual uses, all five of these forms — the preterite indicative, the imperfect indicative and the imperfect subjunctive on the one hand and the pluperfect indicative and the pluperfect subjunctive on the other hand — **temporally locate situations relative to a reference time situated prior to the speech time**. This is the function of the **absolute tense feature [+past]**. Based on this shared temporal anchoring, it is the pluperfect indicative and subjunctive that locate situations prior to that past reference time [+past], whereas the preterite indicative, along with the imperfect indicative and subjunctive, locate them as simultaneous with, or temporally inclusive of, the reference time [+past]. This constitutes the opposition between the **situation-time feature values [+anterior] and [–anterior]**.

We shall now examine the temporal relation between the non-past tenses and the past tenses:

(98) *Pedro es médico*. "Pedro is a doctor."

(99) *Pedro era médico cuando le encontré*. "Pedro was a doctor when I met him."

(100) *Ana ya me ha dicho la verdad*. "Ana has already told me the truth."

(101) *Ana ya me había dicho la verdad cuando le dijiste que no me la dijera*. "Ana had already told me the truth when you told her not to tell it to me."

In (98), the state *Pedro ser doctor* is presented by the present indicative as simultaneous with the speech time and is thus marked as [–past, –anterior], while in (99) = (92) the same state is

presented by the imperfect indicative as simultaneous with a past reference time denoted by the temporal clause *cuando le encontré* and is therefore marked as [+past, –anterior], as shown above. In (100), on the other hand, the event *Ana me decir la verdad* is presented by the present perfect indicative as prior to the speech time and is hence marked as [–past, +anterior], whereas in (101), the same event is presented by the pluperfect indicative as prior to a past reference time expressed by the clause *cuando le dijiste que no me la dijera* and is thus marked as [+past, +anterior].

The feature combination [$\pm$ past, $\pm$ anterior] does not include any feature related to futurity — that is, it does not specify whether a situation is posterior to the reference time or not. Moreover, as we have seen above, the subjunctive tenses do not indicate the futurity status of a situation, which supports the analysis that [$\pm$ past, $\pm$ anterior] are the features that encode temporal anchoring in the subjunctive mood. Taking this into consideration and based on the entire analysis above, we obtain the future-specified version (102) from (82):

(102)

Indicative		Subjunctive	
Present	Future	Present	[–past, –anterior]
Present perfect	Future perfect	Present perfect	[–past, +anterior]
Imperfect	Past future	Imperfect	[+past, –anterior]
Preterite			
Pluperfect	Past future perfect	Pluperfect	[+past, +anterior]

Since, as shown in (83) – (86), both the present perfect and the future perfect indicatives as a whole correspond temporally to the present perfective — specified as [–past, +anterior] — they are likewise specified as [–past, +anterior]. Then, the opposition between the present perfect and future perfect indicatives which share this same feature specification [–past, +anterior] can be analysed either as an opposition in the relative tense feature [$\pm$ posterior], or as an opposition in the modality feature [$\pm$ assumptive] — that is, [$\pm$ doxastically possible]. Since Spanish is not as strongly temporally driven as French with respect to its future tense, which analysis is more appropriate remains an open question. We shall return to this point later.

It is to be noted that "anteriority" does not necessarily imply "finishedness", but "finishedness" (as the cause) necessarily entails "anteriority" (as the effect). Based on this latter entailment and causal relation, the feature [+anterior] can undergo a metonymic semantic extension to [+finished], as in (103):

(103) metonymy: [+anterior] (*effect*) → [+finished] (*cause*)

Let us cite an illustrative example from Japanese. The language has a compound word *tsuki-kage*, composed of *tsuki* ("moon") and *kage* ("shadow"), which would seem to mean "the shadow of an object caused by moonlight". However, in contemporary Japanese, *tsuki-kage* typically refers not to the shadow, but to the moonlight itself. Thus, we observe a metonymic shift as in (104)¹⁶:

(104) metonymy: shadow (*effect*) → light (*cause*)

The fact that the present perfect, pluperfect, future perfect and past future perfect in the indicative, as well as the present perfect and pluperfect in the subjunctive, can express finishedness is accounted for by this metonymic extension. Why, then, are these forms not to be defined from the outset as [+finished]? The reason is that doing so would require the imperfect subjunctive — opposed to the pluperfect, specified hypothetically as [+finished] — to be defined as [–finished]. However, the imperfect subjunctive corresponds not only to the imperfect indicative — which has a [–finished] value — but also to the preterite indicative — which carries a [+finished] value. If the imperfect subjunctive were defined as [–finished], then the [+finished] value of the preterite indicative would become inexplicable. As shall be shown in Section 5.5, this logical conclusion that the Spanish perfect tenses are primitively specified not as [+finished] but as [+anterior] is supported by a metaphorical extension from [±anterior] to [±past] in counterfactual constructions.

The reason why the imperfect and pluperfect subjunctives in Portuguese are opposed in terms of the aspectual feature [±finished], whereas in Spanish — and in French, which employs a similar system — they are opposed by means of the situation-time feature [±anterior], lies in the difference in the oppositional relations among the terms that structure the respective systems of temporal reference in these two languages. This structural difference manifests itself in the unique semantic reanalysis of the present perfect indicative in Portuguese (Section 5.1), in the decline of the preterite and the parallel aoristic reinterpretation of the present perfect in French and Spanish (Section 5.2), in the choice of temporal-aspectual forms in counterfactual

¹⁶ There also exists a near-synonym, *tsuki-akari*, in which *akari* means "light." This compound straightforwardly denotes "moonlight." Interestingly, however, *tsuki-kage* and *tsuki-akari* are not entirely synonymous. While the latter simply denotes "the brightness produced by moonlight," with no implication of shadows, the former retains an implicit reference to the presence of shadows cast by the moon. For instance, *tsuki-kage no warutsu* ("The Waltz of the Moonlight (and Shadows)") evokes an image of two dancers and their moving shadows illuminated by the moon. By contrast, *tsuki-akari no warutsu* ("The Waltz of the Moonlight") fails to conjure this imagery of shadows altogether. In the **metonymic extension of semantic features** as well, it may be therefore necessary to take into account meaning shifts of the type: *effect* → *cause* (+*effect*).

constructions (Section 5.5), and in the maintenance of the preterite-imperfect opposition in the subjunctive in Portuguese and its neutralisation in Spanish and French (Section 5.6).

Returning to the preterite and the imperfect indicatives, their opposition cannot be captured by the absolute tense feature [past] and the situation-time feature [anterior] alone, since both are characterised as [+past, –anterior]. Moreover, since [+anterior] has been argued to undergo a metonymic extension to [+finished] in all the perfect forms, it follows that the preterite indicative, specified as [–anterior], cannot extend this feature to [+finished] either. This is where the notion of the speaker's temporal perspective comes into play. The imperfect indicative is a tense form in which the speaker's temporal viewpoint is placed on the reference time used for situation anchoring — that is, on [+past, –anterior], which logically reduces to [+past]. By contrast, the preterite indicative places the speaker's viewpoint not on the reference time [+past], but on [–past] — more precisely, on the default value of [–past], namely the speech time.

The opposition between the *finished* aspect of the preterite and the *non-finished* aspect of the imperfect is derived from this difference in the location of the speaker's viewpoint. In the imperfect, the temporal viewpoint overlaps with the situation time, making it impossible for the speaker to confirm whether the situation has finished, as shown in (105) = (92):

(105) *Pedro era médico cuando le encontré.* "Pedro was a doctor when I met him — it is not specified whether he still is."

(106) *Pedro fue médico.* "Pedro was a doctor — it is specified that he is no longer one."

In the preterite (106) = (91), by contrast, the speaker's viewpoint is external to the situation time, so the ending of the situation is *potentially* observable. It is important to note, however, that while this external viewpoint makes the confirmation of finishedness *possible*, it merely places the speaker in a position to verify it — the situation itself may, in fact, remain unfinished. The preterite's ability to convey a finished aspect thus depends on its minimal semantic opposition to the imperfect. The imperfect, due to the internality of the speaker's viewpoint — that is, the non-verifiability of finishedness — must be interpreted as [–finished] and the preterite, as its counterpart in the minimal semantic pair, receives a [+finished] interpretation. In other words, aspect here is a derived value.

To represent this opposition, the feature [$\pm VP \in \text{past}$] is introduced. In contrast, the imperfect subjunctive is left unspecified for this feature, which explains why it can be used in both preterite-like and imperfect-like contexts. Moreover, the contrast between the preterite and the imperfect is unique within the system: they are the only two forms that share the same values

for [non-assertive], [past] and [anterior]. Consequently, no other forms in the system are specified for [$\pm VP \in \text{past}$], since there is no formal opposition that would warrant such specification.

In systems where temporal location is determined on the basis of the relation to the speech time as a reference time, the distinction between simultaneity with that point — "present" — and non-simultaneity — "past" and "future" — is a fundamental one. The reason is straightforward: the temporal relation to the reference time — namely, the speech time — differs. At the same time, "present" and "future" stand in opposition to "past" as a unified category — that of "non-past". The reason for this opposition is equally clear: the PAST is characterised by the impossibility of intervention in the situations that belong to it, whereas the NON-PAST is characterised by the possibility of intervention in its constituent situations. The two categories are thus ontologically distinct. This constitutes the theoretical foundation for introducing the opposition between the PAST and the NON-PAST into the Portuguese perspective feature [$\pm VP \in \text{past}$]. Moreover, in Portuguese, this opposition is formally manifested in the contrast between the viewpoints of the imperfect/pluperfect and those of the present/preterite.

In this way, while "past" is equivalent to "non-present" and thus formally symmetrical with "future", it is not equivalent to "future", insofar as the latter is grouped together with "present" to form the category "non-past", which stands in opposition to "past". This compels us to posit that the structuring of time with reference to the speech time necessarily involves a fundamental opposition between "past" and "non-past (= present + future)". That is, even in the structuring of absolute tense, it may be more appropriate not to assume an immediate three-way opposition of "past / present / future" from the outset, but rather to posit two successive binary oppositions: "past" vs "non-past" and, within "non-past", "present" vs "future".

Indeed, Japanese exhibits only the opposition between the "past" *-ta* and the "non-past" *-ru*. In English, too, it is the presence or absence of *-ed* that marks the opposition between "past" and "non-past", and the "non-past" category is further divided into "present" and "future" through the presence or absence of the modal auxiliary *will*. English also allows the internal structure of the "past" category to be further articulated by the use of modals such as *would* — the past form of *will*. In the case of Spanish as well, the decision to characterise absolute tense primarily in terms of the feature [$\pm \text{past}$], rather than positing a three-way opposition including an independent "future" value, is based on the same line of reasoning.

On the basis of this, the reason for defining future tense forms not in terms of a tense feature [+posterior], but rather through the modality feature [+assumptive] — namely, [+doxastically

possible] — lies in the fact that the semantic feature that characterises them cannot be unambiguously classified as a "pure tense feature", unlike [$\pm$ past] or [$\pm$ anterior]. In Spanish, the function of the future tense is not simply to anchor a situation in future time. Rather, it expresses the speaker's belief that a future situation is likely to occur — a representation of the speaker's cognitive belief system. It is precisely this function that allows the future tense, originally used for expressions of assumption, to extend naturally to expressions of intention or command. For this reason, it is difficult to treat what the Spanish future tense expresses as being purely a matter of the tense feature [$\pm$ posterior]. Its core meaning lies in doxastic possibility — that is, in the representation of a belief-based prediction.

(*Doxastic possibility* refers to a possibility that holds within the speaker's belief system. That is, it expresses the speaker's belief in the possibility of the occurrence of a situation. It corresponds not to epistemic "may", but to epistemic "will". To speak about future situations entirely in terms of "may" is ultimately meaningless, as "may" implies a 50/50 possibility — i.e., the speaker conveys nothing definite about the future. To speak about the future is, rather, to speak about what one believes is likely to happen.)

From this viewpoint, the oppositions in Spanish between present indicative and future indicative, between preterite/imperfect and past future, and between pluperfect and past future perfect, can be seen — just as in Portuguese — as expressing [$\pm$ doxastically possible], or [$\pm$ assumptive], features interpretable both as temporal and modal. The same applies to the opposition between the indicative and subjunctive moods. This, too, expresses a modal contrast — specifically, [$\pm$ suspension-of-truth-value-judgment/commitment], or, equivalently, [$\pm$ non-assertive].

The feature analysis outlined above is summarised in tabular form in (107). It corresponds substantially to (9):

(107)

[−non-assertive]		[+non-assertive]	
Present	Future	Present	[−past, −anterior]
Present perfect	Future perfect	Present perfect	[−past, +anterior]
Imperfect [+VP $\in$ past]	Past future	Imperfect	[+past, −anterior]
Preterite [−VP $\in$ past]			
Pluperfect	Past future perfect	Pluperfect	[+past, +anterior]
[−assumptive]	[+assumptive]	[0 assumptive]	

By contrast, the system of temporal anchoring in contemporary standard spoken and written French is shown in (108). It corresponds to (107) with the preterite indicative and the imperfect and pluperfect subjunctives removed, and substantially corresponds to (13):

(108)

[−non-assertive]		[+non-assertive]	
Present	Future	Present	[−past, −anterior]
Present perfect	Future perfect	Present perfect	[−past, +anterior]
Imperfect	Past future	—	[+past, −anterior]
Pluperfect	Past future perfect	—	[+past, +anterior]
[−posterior]	[+posterior]	[0 posterior]	

5. Theoretical Applications

On the basis of the structural differences between the perspective-and-aspect-driven temporal anchoring system and the tense-driven one, we shall account for some striking contrasts between Portuguese and other major Romance languages — especially Spanish and French.

5.1 The Systemically Driven Semantic Shift of the Portuguese Present Perfect

Whereas the present perfect (*pretérito perfecto compuesto*) in Spanish and the compound past (*passé composé*) in French might express the original present perfect value — namely, the termination of a situation prior to the speech time and the continuation of its result state up to that moment (i.e. resultative or experiential readings) — the Portuguese present perfect indicative (*pretérito perfeito composto do indicativo*, PPC: *ter* in the present indicative + the unmarked form of the past participle — masculine singular), which shares the same historical origin, has today a markedly different temporal value. In contemporary Portuguese, it denotes the initiation of a continuous or iterative state and its non-termination at the speech time. **This semantic reanalysis is not arbitrary**, but results from the fact that Portuguese employs a perspective-and-**aspect**-driven temporal system — in which the temporal anchoring is completed through a two-stage process: $[\pm VP \in \text{past}] \Rightarrow [\pm \text{finished}]$ — in contrast to the tense-driven systems of its Romance counterparts — where the temporal anchoring is completed through a process of up to four stages: $[\pm \text{past}] \Rightarrow [\pm \text{posterior}] \Rightarrow [\pm \text{anterior}] (\Rightarrow [\pm VP \in \text{past}])$.

The Portuguese present perfect originally took the form of the possessive verb *haver* in the

present indicative followed by a noun phrase (NP) denoting the possessed object and a transitive past participle (pp.) in agreement with that NP in gender and number (e.g. *Hei um livro lido* 'I have a book written'). Semantically, it conveyed the meaning "to have something done" — analytically decomposable as:

- (i) the termination of an event prior to the speech time;
- (ii) the initiation of its result state that begins simultaneously with that termination (i);
- (iii) the continuation of the result state up to the speech time;
- (iv) a possessive relation; and
- (v) the notion of a possessed object.

Morphologically, these semantic components were encoded as follows (here, [$\sqrt{\text{theme}}$] represents the theta-role assigned by a transitive verb):

- (i) was encoded by the past participle as [+finished] and by the present indicative on *haver* as [$-\text{VP}\in\text{past}$]
- (ii) was also encoded by the past participle as [$\sqrt{\text{initiated}}$, $\sqrt{\text{result-stative}}$].
- (iii) was encoded by the past participle as [$\sqrt{\text{initiated}}$, $\sqrt{\text{result-stative}}$] and by the present indicative on *haver* as [$-\text{VP}\in\text{past}$, $-\text{finished}$].
- (iv) was encoded by the radical of the verb *haver* as [$\sqrt{\text{possessive}}$].
- (v) was encoded by the NP as [$\sqrt{\text{theme}}$, $\sqrt{\text{possessed}}$].

(As will be discussed in Section 5.7, it remains indeterminate at the present stage of our research whether, at the point when the original form of the PPC emerged, the Classical Latin tense feature [$\pm\text{past}$] had already shifted to the perspective feature [$\pm\text{VP}\in\text{past}$], characteristic of Modern Portuguese. However, since in the reanalysis of the PPC **the crucial factor was the aspectual feature [+finished]** and the distinction between [$\pm\text{past}$] and [$\pm\text{VP}\in\text{past}$] is assumed to have been of relatively little importance, we shall employ the notation [$\pm\text{VP}\in\text{past}$] for convenience.)

In terms of the syntactic constituent, *haver* in the present indicative, the past participle, and the NP bear the following lexical and functional features (here, L or F indicates whether a given feature is lexical or functional, [$\sqrt{\text{theme}}$] globally represents the theta-role assigned by a transitive verb, and *pp.* is an abbreviation of past participle):

- *haver*: L [$\checkmark$ possessive] / F [$\neg$ VP $\in$ past, $\neg$ finished]
- pp.: F [+finished [$\checkmark$ initiated, $\checkmark$ result-stative]]
- NP: L [$\checkmark$ theme, $\checkmark$ possessed]

In Spanish, French and Italian, which use the auxiliaries *haber*, *avoir* and *avere* — all of which originated from the Latin verb *habēre*, like the Portuguese *haver* — the semantic origin is assumed to be the same across all these languages.

However, due to syntactic restructuring — on the surface, the rightward movement of the noun phrase and the loss of gender and number agreement on the past participle — the semantic components of (iv) "possession" and (v) "possessed object" were eliminated from the original syntactic-semantic construction as follows:

- *haver*: F [$\neg$ VP $\in$ past, $\neg$ finished]
- pp.: F [+finished [$\checkmark$ initiated, $\checkmark$ result-stative]]
- NP: L [$\checkmark$ theme]

Syntactically and semantically, this change involved the noun phrase shifting from being the complement of *haver* to the complement of the past participle. This shift entailed the loss of its interpretation as referring to a possessed object, as well as the loss of *haver*'s function as a possessive verb. At the same time, the noun phrase's relocation to the complement position of the past participle gave rise to a verb phrase headed by the participle, thereby resulting in the disappearance of gender and number agreement. As the past participle came to appear immediately after *haver*, a new phrase structure emerged in which *haver* selects as its complement the entire verb phrase headed by the past participle. As a result, the newly reanalysed structure *haver* + past participle in the masculine singular form + noun phrase came to express the meaning of (i) the termination of a situation prior to the speech time, and (ii) the initiation and continuation — up to the speech time — of the result state that begins simultaneously with that termination. (Since the termination point of the situation precedes the speech time, there exists a temporal interval between these two points. Moreover, because the termination of the situation coincides with the initiation of its result state, there is also an interval between the beginning of this state and the speech time — and this interval corresponds to the duration or continuation of the result state.) Subsequently, as this structure expanded to include intransitive verbs — in this case, however, the verb *ser* in the present indicative +

variable past participle was a preferred construction — the transitive relation between the verb and its argument was further eliminated, as follows:

- *haver*: [−VP∈past, −finished]
- pp.: [+finished [√initiated, √result-stative]]

"In the 15th century, *ter* begins to occur as an expression of unmarked possession, thus competing with *haver* as a lexical verb. *Haver* becomes marginal in the 16th century and is definitively obsolete from the 17th century onward. In parallel with its lexical development, *ter* begins to substitute *haver* in the perfect periphrasis." (cf. Olbertz 2018: 486):

- *ter*: [−VP∈past, −finished]
- pp.: [+finished [√initiated, √result-stative]]

In Spanish, French and Italian, on the other hand, a similar process gave rise to the *pretérito perfecto compuesto*, *passé composé* and *passato prossimo*, respectively. However, since these languages operate with tense-driven temporal anchoring systems, such constructions came to be specified as [−past, +finished, [√initiated, √result-stative]], [−past, +finished] or [−past, +anterior], and stabilised within the system as semantically distinct counterparts to the preterite, which is specified as [+past, −anterior, −VP∈past]¹⁷.

Portuguese operated, in contrast, with a perspective-and-aspect-driven system, and the situation is therefore completely different. As discussed in Section 2.1, in the domain of fact-stating forms marked as [−non-assertive, −assumptive], the only primitive anchoring-aspect features used for temporal anchoring are [±VP∈past] and [±finished].

The four possible combinations of these features are exhaustively and exclusively realised by the *present* [−VP∈past, −finished], *preterite* [−VP∈past, +finished], *imperfect* [+VP∈past, −finished] and *pluperfect* [+VP∈past, +finished] of the indicative, respectively. The condition that this core paradigm — which appears to have remained essentially unchanged since Classical Latin, in that it formally encodes the feature values [±finished], comprising the quasi-

¹⁷ Conversely, in the course of the emergence of the present perfect, it is also possible that the preterite, originally specified as [−past, +anterior], was displaced into the domain of [+past, −**anterior**], where the imperfect resided. Such a displacement would have been impossible, however, if that domain were primitively [+past, −**finished**], for the preterite — through a metonymic extension from [+anterior] to [+finished] — was capable of encoding [+finished] as well. It was precisely because the target domain was [+past, −anterior], which does not primitively encode aspectual values, that the preterite could relocate there and come to be opposed to the imperfect, whose perspective feature is [+VP∈past], via the perspective feature [−VP∈past] derived from its original [−past] specification.

equivalents of *præsens* [–past, –finished], *perfectum* [–past, +finished], *imperfectum* [+past, –finished] and *plusquamperfectum* [+past, +finished] of the indicative — is already exhaustively saturated results in a structurally closed system, leaving no room for additional forms. It is this structural closure that triggered the semantic shift of the present perfect indicative. Let us now examine this development in detail.

As noted earlier, the original temporal value of the present perfect in the indicative mood was the combination of (i) the termination of a situation prior to the speech time and (ii) the initiation and continuation up to the speech time of a result state beginning simultaneously with that termination. Within this system, however, the preterite indicative, specified as [–VP∈past, +finished], had long functioned as the highly stable form responsible for expressing the termination of a situation prior to the speech time. (

Moreover, this tense was capable — depending on context or situation and especially in combination with temporal adverbials such as *já*, *nunca* or *ainda não* — of expressing not only the termination of the situation itself, but also the initiation, continuation and non-termination at the speech time of its result state. Within such a system, the newly formed construction *ter* in the present indicative + past participle had no room to compete with the preterite indicative, which was already capable of fulfilling the same temporal function. As a result, although the present perfect indicative originally bore the prototypical present perfect value — [–VP∈past, +finished [✓initiated, ✓result-stative, –finished]] — its temporal value underwent a shift. It came to be reinterpreted as a form marked by [–VP∈past, ✓initiated, ✓stative, –finished], and survived in the system in opposition to the preterite indicative, specified as [–VP∈past, +finished], and the present indicative, specified as [–VP∈past, –finished].

This process may be hypothesised as follows. As has already been stated on several occasions, the original temporal interpretation of the present perfect indicative comprised:

- (i) the termination of a situation prior to the speech time, and;
- (ii) the initiation — simultaneous with (i) — of a result state characterised by:
 - (a) its beginning prior to the speech time;
 - (b) its continuation up to that point, and;
 - (c) its non-termination at the speech time.

In a perspective-and-aspect-driven system, this configuration corresponds to the feature specification [–VP∈past, +finished [✓initiated, ✓result-stative, –finished]] — where the

complex feature combination [+finished [√initiated, √result-stative]] is encoded by the past participle and [−VP∈past, −finished] is encoded by the present indicative on *ter*.

Of the two components, (i) was encoded by the feature combination [−VP∈past, +finished], where the value [+finished] entered into direct competition with the same value exhibited by the preterite indicative — likewise specified as [−VP∈past, +finished] — which already occupied a stable and dominant position within the [±VP∈past] × [±finished] system. Consequently, the present perfect came to **relinquish the feature [+finished] itself** in favour of the preterite indicative. The other component, (ii) *the initiation — simultaneous with (i) — of a result state characterised by its beginning prior to the speech time, its continuation up to that point, and its non-termination at the speech time*, represents the distinctive value of the present perfect, namely [√initiated, √result-stative, −finished]. However, since (ii) constituted a temporal structure that inherently presupposed (i), it could not be used in isolation. As a result, through a process of synecdochic extension, the interpretation shifted to the more general meaning of *the initiation — simultaneous with (i) — of a result state characterised by its beginning prior to the speech time, its continuation up to that point, and its non-termination at the speech time*, and this reinterpretation — namely, [+VP∈past, √initiated, √stative, −finished] in terms of feature reconfiguration — came to be applied to the description of a wide range of situations¹⁸. The feature reconfiguration appears to have proceeded as follows:

- *ter*: [−VP∈past, −finished]
- *pp*.: [+finished [√initiated, √result-stative]]

The feature [+finished] was then deleted from the past participle, and, simultaneously, by means of synecdoche, the [result-] component was removed from [√result-stative].

- *ter*: [−VP∈past, −finished]
- *pp*.: [~~+finished~~ [√initiated, √~~result~~-stative]]

The final result was as follows:

- *ter*: [−VP∈past, −finished]

¹⁸ In connection with this semantic shift, the past participle in the present perfect indicative also came to denote not its original meanings of "termination" or "completion", but rather "partial realisation". The past participle's original function — to express the termination of a situation as viewed from a particular point in time — underwent a metonymic shift to the complete realisation of the situation. This, in turn, was metaphorically extended, via the schema of "realisation", to denote the partial realisation of the situation.

- pp.: [✓initiated, ✓stative]

Ultimately, the enigma of the Portuguese present perfect indicative lies in the semantic reanalysis of the participle itself: from a [+finished [✓initiated, ✓result-stative]] participle to a [✓initiated, ✓stative] participle. The features [−VP∈past, −finished] on the auxiliary *ter* remained virtually unchanged throughout. Accordingly, all variation in the meanings of perfect constructions can be traced back to this strikingly singular shift in participle semantics. In fact, in Portuguese, all compound temporal-reference forms except *tem falado* — namely, *tinha falado*, *terá falado*, *teria falado*, *tenha falado*, *tivesse falado*, and *tiver falado* — are specified as [+finished]. It follows, then, that **in the past participles of these forms, a shift occurred from [+finished [✓initiated, ✓result-stative]] to [+finished], by discarding the component of result-state transition** — in contrast to the present perfect indicative, which retains it. These perfect forms, however, differ from the PPC₂ in that the [−finished] component originally encoded by the auxiliary — specified as [−VP∈past, −finished] or [−VP∈past, −finished] — has been eliminated. For instance, in the case of the pluperfect indicative *tinha falado*, the feature configuration undergoes the following shift:

[+VP∈past, +finished [✓initiated, ✓result-stative, −finished]]

↓

[+VP∈past, +finished [~~✓initiated~~, ~~✓result-stative~~, ~~−finished~~]]

↓

[+VP∈past, +finished]

In this process, the compound pluperfect relinquished the [−finished] feature originally present in the auxiliary *tinha*, in order to signal only the [+finished] aspect.

Why, then, did the compound present perfect form not replace the simple preterite form in Portuguese, as it did in other Romance languages? The answer lies in the fact that the form that appeared to be a *preterite* — the PPS — was in fact, as we already know, a *present perfect* form, with "perfect" here denoting finishedness without reference to a result state, as in the Classical Latin *perfectum* in its primary sense.

If we designate as PPC₁ the earlier stage in which the present perfect construction expressed the termination of an event prior to the speech time and the persistence of its result state, and as PPC₂ the later stage in which it came to denote the initiation and continuation of a state prior to the speech time as well as its non-termination at that point, then the shift between these two

states can be described as involving the following changes:

PPC₁ → PPC₂

[−VP∈past, +finished [√initiated, √result-stative, −finished]]

↓

[−VP∈past, ~~+finished~~ [√initiated, ~~√result-stative~~, −finished]]

↓

[−VP∈past, √initiated, √stative, −finished]

The fact that the present perfect indicative still expresses the value [√stative] finds its origin in this historical development. Consequently, it is naturally compatible with predicates whose *Actionsart* is stative, and straightforwardly presents them — depending, however, on the context, including adverbials, the *Aktionsart* of the predicate, and encyclopedic knowledge (as will be discussed later) — as situations characterised by an initial point, temporal duration, and the absence of a terminal point at the speech time, as shown in (109):

(109) *Nasci em Coimbra e tenho vivido cá desde então.* "translation"

In (109), the *Aktionsart* of the verb *viver* is stative, and is therefore compatible with the feature [√stative] of PPC₂. The features [−VP∈past, √initiated] mark this state as having an initiation point prior to the speech time, which coincides with *então*, itself anaphoric to (*o momento em que*) *nasci em Coimbra*. The fact that the speaker still lives in Coimbra at the speech time is denoted by the feature combination [−VP∈past, −finished].

By contrast, in the case of non-stative predicates such as processes, achievements, accomplishments and semelfactives — that is, events in general — the interpretation involves first reanalysing them as iterative states: repeated events construed as states. An event consists of a sequence of distinct phases and therefore does not permit homogeneous continuity: in other words, unlike a state, a part does not equal the whole. To homogenise — that is, to stativise — it, there is no means other than repetition. This is because any portion of a repetition still counts as a repetition. They are then presented as [−VP∈past, √initiated, √stative, −finished]. Thus, even in the absence of temporal adverbials, the interpretation of present perfect forms as indicating *iteration* in the case of eventive predicates derives from the aspectual operations described above.

However, a difficult problem arises in the case of stative predicates:

(110) *Esta manhã tem chovido muito.* "This morning it has been raining a lot." (continuity)

(111) *Tem chovido muito.* "It has been raining a lot repeatedly." (repetition)

(112) *Ultimamente tem chovido muito.* "Lately it has been raining a lot repeatedly." (repetition)

In (110), the rain that began to fall this morning has continued to fall and is still falling at the speech time, which is itself still located within the same morning. This continuity interpretation appears to derive directly from the feature combination $[-VP \in \text{past}, \checkmark \text{initiated}, \checkmark \text{stative}, -\text{finished}]$. In (111) and (112), nevertheless, this continuous reading is no longer available. Instead, the situations are interpreted as iterative states composed of an unspecified number of finished states. Where does this reading come from? We hypothesise that the iterative stative interpretation of eventive situations, which is coerced by the $[\checkmark \text{stative}]$ feature inherent in the present perfect indicative, had been habitually performed; as a result, a temporal and aspectual cognitive frame — that of "the initiation and continuation of an iterative state from a certain point in the PAST up to the speech time" — is established and comes to be associated with the present perfect indicative, thereby inducing an iterative stative interpretation even for stative situations.

As for stative situations, the feature combination $[-VP \in \text{past}, \checkmark \text{initiated}, \checkmark \text{stative}, -\text{finished}]$ gives rise to an interpretation in which a state is understood as having begun prior to the speech time, continued up to that point, and remained unfinished at the speech time. This specification, however, does not determine whether the state is interpreted as iterative or continuous. At this point, the temporal and aspectual cognitive frame described above intervenes — namely, the continuation of an iterative state from a certain point in the past up to the speech time. In (97), which lacks any temporal adverbial, the iterative reading derives from this frame. In (98), which includes a temporal frame compatible with it, the iterative reading likewise arises. In (96), by contrast, the temporal frame is incompatible with this cognitive frame; the frame is thus neutralised, and the continuous reading prevails.

Additionally, the present perfect indicative is incompatible with temporal adverbials that explicitly quantify the number of iterations. This is because such quantification presupposes a time interval that is closed at both ends: counting events requires not only a defined starting point but also a defined endpoint. The present perfect, by its very nature, denotes a time interval that is open at the endpoint, thereby resulting in a semantic mismatch.

Besides the commonly discussed uses mentioned above, the present perfect indicative also exhibits an interesting additional usage. This usage may be described as the temporal anchoring

of a composite state, composed of two sequential states, by means of the present perfect. Araújo (2003: 232–233) characterises it as follows: "L'événement est **terminé** mais il est **mis en relation avec un autre qui le suit**. Dans ce cas *le pretérito perfeito composto* insiste sur la continuité ou la répétition de cet événement (à la différence du *pretérito perfeito simples*)". On p. 233 of the same work, Araújo presents the following two examples:

(113) *O doente tem estado muito agitado mas agora acalmou.* "The patient has been very agitated, but he has now calmed down."

(114) *Têm tomado várias decisões importantes, mas atualmente evitam assumir novas responsabilidades.* "They have been making several important decisions, but at present they avoid taking on new responsibilities."

At first glance, the fact that the first situation can be viewed as finished from the speech time might seem to contradict the earlier characterisation, but in fact it does not. In (113), the state₁ *o doente estar muito agitado*, together with the subsequent state₂ *o doente estar calmo*, forms a single composite state₃, which is marked as [$\sqrt{\text{stative}}$]: state₃ = $\langle \text{state}_1 \cup \text{state}_2 \rangle$. The initiation of this composite state₃ — which coincides with that of state₁ — is denoted by the feature [$\sqrt{\text{initiated}}$], and its non-termination at the speech time — which coincides with that of state₂ — is denoted by the feature combination [$-\text{VP} \in \text{past}$, $-\text{finished}$]. The transition from state₁ to state₂ within the composite state₃ is encoded lexically by *acalmar*. Analytically, the sequence can be represented as follows: initiation of state₁ $\Rightarrow$ continuation of state₁ $\Rightarrow$ transition from state₁ to state₂ $\Rightarrow$ non-termination of state₂ at the speech time. The present perfect's feature combination [$-\text{VP} \in \text{past}$, $\sqrt{\text{initiated}}$, $\sqrt{\text{stative}}$, $-\text{finished}$], applied to the whole composite situation₃, is morphologically realised on the first predicate. What is crucial here is that the combination [$-\text{VP} \in \text{past}$, $-\text{finished}$] indicates that the situation₃ as a whole is not finished at the speech time. It does not mean, on the contrary, that state₁ itself remains unfinished — the feature [$-\text{finished}$] does not denote the termination status of the state₁ referred to by the predicate *estar muito agitado* on which this feature is realised. Even if state₁ has come to an end — that is, even though a transition to state₂ has occurred — the above feature combination still clearly presents the composite state₃ as ongoing at the speech time. In (114), too, the repeated action *tomar decisões* construed as the state₁ and the subsequent absence of the action *assumir novas responsabilidades* — that is, the subsequent state₂ — together constitute a single composite state₃. Analytically, the sequence of states may be represented as: initiation of the state₁ *tomar decisões* $\Rightarrow$ continuation of state₁ $\Rightarrow$ transition from state₁ to state₂ *não assumir novas responsabilidades* $\Rightarrow$ non-termination of state₂ at the speech time. Accordingly, the situation is

still being described in terms of $[-VP\in past, \checkmark initiated, \checkmark stative, -finished]$.

There are other cases that are subject to complex analysis. In contexts such as live news reporting, it may appear that the present perfect indicative is incompatible with an achievement predicate like *registar um aumento de casos nos últimos dias*, as in (115):

- (115) *Temos registado um aumento de casos nos últimos dias*. "We have registered an increase in cases in recent days."

However, the present perfect indicative as a form functions here by reinterpreting the event *registar* as an iterative state — one that was initiated prior to the speech time and continues up to it. The phrase *um aumento de casos nos últimos dias*, on the other hand, serves to focus on one portion of that ongoing state — that is, it isolates one segment of the whole. Still, since "a part of a state" is itself a state, the focused portion remains compatible with the stative value of the present perfect.

As has been said in Section 2.5, when **the temporal viewpoint is not located exclusively on the speech time**, the present perfect indicative may have **the same temporal value as the preterite indicative — namely, $[+VP\in past, +finished]$** — as exemplified in (116) = (73).

- (116) *Sempre que a Ana chega a casa da Maria, já o Rui a tem visitado / visitou*. "Whenever Ana arrives at Maria's house, Rui has already visited her" (from Mateus et al. 2003: 143).

This type of viewpoint placement is far less common than its canonical anchoring to the speech time alone, and it is possible that the present perfect indicative has not historically been perceived as functionally competing with the preterite indicative. Accordingly, in this usage, the specification of the past participle has not shifted from $[+finished [\checkmark initiated, \checkmark result-stative]]$ to $[\checkmark initiated, \checkmark stative]$, but rather to $[+finished]$, as in the other compound tense forms.

Finally, although not pursued in depth here, we propose that the Portuguese present perfect may alternatively be analysed as the present indicative specified as $[-non-assertive, -VP\in past, -finished, -assumptive]$, plus a dedicated operator "*ter* + past participle" encoding $[\checkmark initiated, \checkmark stative]$. This hypothesis, probably first advanced in the present study, would account for the continuative or iterative readings by coercing eventive predicates into an iterative state profile. A systematic corpus-based test of stative verbs is, however, left for future work.

5.2 Systemic Necessity of the Present Perfect's Aoristic Drift and the Decline of the Preterite in French and Spanish

Building on the structural difference between tense-driven and perspective-and-aspect-driven temporal anchoring systems — which served as the starting point for our analysis of the semantic singularities of the Portuguese present perfect in Section 5.1 — this section begins by examining how Spanish distinguishes between the preterite indicative and the present perfect indicative in expressing past situations that are viewed as finished. In Spanish, situations that are viewed as finished within what may be called the "psychological present" — that is, a temporally extended interval centred on the speech time, such as *hoy, esta semana, este mes, este año, esta década*, or *este siglo* — are normally expressed using the present perfect indicative, as in (117). By contrast, situations that are viewed as finished outside of the psychological present — e.g. *ayer, la semana pasada, el mes pasado, el año pasado, la década pasada*, or *el siglo pasado* — are generally expressed using the preterite indicative, as in (118):

(117) *He ido al banco hoy / esta semana / este mes.* "I went to the bank today / this week / this month."

(118) *Fui al banco ayer / la semana pasada / el mes pasado.* "I went to the bank yesterday / last week / last month."

If we divide the time axis with reference to the speech time into an interval that excludes the speech time and extends backward from it, and an interval that includes the speech time and extends forward, then the former corresponds to the PAST and the latter to the NON-PAST. The critical issue lies in determining the boundary between the PAST and the NON-PAST — a boundary that is variable under the condition that the NON-PAST must always include the speech time. For example, when the present indicative is used to express a *habitual present*, the boundary between PAST and NON-PAST can shift significantly backward from the speech time — reaching into what would ordinarily be considered the past in everyday terms. The notion of the *psychological present* also belongs to this type of NON-PAST. It refers to a temporally extended interval centred on the speech time, and it is interpreted as NON-PAST precisely because it necessarily includes the speech time. Accordingly, a situation that is finished within this NON-PAST interval must be expressed by a form marked as [–past], and, since it is finished, it must also be expressed by a form that bears [+anterior] — a feature that, as discussed in Section 4, can undergo a metonymic shift to [+finished]. The only form that satisfies these conditions within the [–non-assertive] domain is the present perfect indicative, specified as [–past, –assumptive, +anterior]. This semantic extension of features is represented in (119):

(119) [–past, –assumptive, +**anterior**] → [–past, –posterior, +**finished**]

In opposition, the "past outside the psychological present" excludes the speech time, and thus constitutes a genuine PAST. Consequently, any situation that is finished within this interval must be expressed by a form marked as [+past] and [–anterior] — since [+anterior] would locate the situation even further back in time, beyond the past interval within which it is situated — and simultaneously marked as [+finished]. The only form that satisfies these conditions within the [–non-assertive] domain is the preterite indicative, specified as [+past, –assumptive, –anterior, –VP∈past], where the feature [–VP∈past] is interpreted as [+finished] via metonymic extension (see Section 4). This feature extension is shown in (120):

(120) [+past, –assumptive, –anterior, –**VP∈past**] → [+past, –posterior, –anterior, +**finished**]

By contrast, such a distinction is not possible in Portuguese, due to the nature of its temporal anchoring system. Since the language lacks the absolute tense feature [±past] as a grammatical category, there is no means of distinguishing between a situation that is finished within the psychological present and one that is finished outside of it. Both are uniformly expressed by the preterite indicative specified as [–VP∈past, +finished]. It is not possible, on the other hand, to express a situation that is viewed as finished in the PAST outside the psychological present by using the speaker's viewpoint located within that same PAST interval — namely, [+VP∈past]. This is because denoting the finishedness of the situation requires the aspectual feature [+finished], but the combination of [+finished] with [+VP∈past], which is encoded by the pluperfect indicative, would locate the situation even further back in time — prior to the PAST outside the psychological present. The imperfect, specified as [+VP∈past, –finished], is also unsuitable, since it cannot present the situation as finished. Accordingly, irrespective of when the situation actually finished, any situation that is viewed from the speech time as finished is necessarily expressed by the preterite indicative.

On the other hand, in French — where the simple past has long disappeared from spoken usage and is no longer used in everyday written language either — it is no longer possible to distinguish between a situation ending within the psychological present and one ending in the past outside of it by means of the compound past (*passé composé*) versus the simple past (*passé simple*). However, French did historically make such a distinction, just as Spanish does today (cf. "règle des vingt-quatre heures" in Krell 1987: 368). The current state of Spanish, therefore, reflects an earlier stage once exhibited by French itself. The French compound past, which originally emerged to express the present perfect value — that is, the termination of a situation prior to the speech time and the continuation of the result state up to that moment — later came

to be used for situations that ended within the *psychological present*. Eventually, it was further extended to denote any situation finished before the speech time. This expansion of the compound past's function is, in our view, most plausibly related to the development of the narrative present (*présent historique*) usage of the present indicative.

The narrative present is based on a temporal interval constructed by sliding the boundary between [+past] and [−past] as far leftward as possible — under the condition that the speech time must always remain within the [−past] interval. As a result, the linguistic [−past] (i.e., the non-past domain) becomes extensible into what is ordinarily regarded as the past in everyday terms. Consequently, a situation that ended in this expanded interval can now be encoded by a [−past] form, insofar as it is viewed as having finished within that [−past] interval. The remaining question is how such finishedness is linguistically encoded. This becomes possible if the relative-tense feature [+anterior] undergoes a metonymic shift to [+finished], as previously argued. The compound past in French, specified as [−past, +anterior], is precisely the form that satisfies this condition. The above historical process can be modelled as follows: the establishment of the narrative present leads to the leftward expansion of the non-past interval, which in turn increases the functional scope of the compound past.

The simple past, in parallel, underwent a narrowing of its functional domain — from generally marking situations finished before the speech time to marking only those finished in the past outside the *psychological present* — and eventually fell into disuse in spoken language, along with the past anterior (*passé antérieur*). This was because these forms were the only ones marked not only as [+past] but also as [−VP∈past] — the simple past (e.g. *il fit* "he did"), specified as [+past, −posterior, −anterior, −VP∈past], and the past anterior (e.g. *il eut fait* "he had done"), as [+past, −posterior, +anterior, −VP∈past]. It seems that a dynamic was set in motion that led to the exclusion of exceptionally highly marked forms. But why exceptionally highly marked? The answer lies in the temporal anchoring process in (121) = (14):

(121)

Temporal anchoring process				
	1 st stage	2 nd stage	3 rd stage	4 th stage
PT	[±VP∈past] speaker's temporal viewpoint located	[±finished] finishedness evaluated → situation time located	[±assumptive] occurrence possibility evaluated	
SP	[±past] reference times defined	[±posterior]←[±assumptive] sub-reference times defined	[±anterior] situation time located	[±VP∈past] viewpoint selected
FR	[±past] reference times defined	[±posterior] sub-reference times defined	[±anterior] situation time located	[±VP∈past] viewpoint selected

PT, SP, and FR stand for Portuguese, Spanish and French, respectively.

In French — as in Spanish — the 'viewpoint past' feature [$\pm VP \in \text{past}$] is only applied at the fourth anchoring stage, long after the core past vs. non-past, posterior vs. non-posterior, and anterior vs. non-anterior oppositions have been established. Furthermore, this late-arriving feature can encode at most two pairs within an already tightly packed tense system. There was no functional niche for additional past-oriented forms such as the simple past and the past anterior.

Also in Spanish, the preterite anterior (e.g. *hubo hecho* "he had done") — which shared substantially the same feature values as the French past anterior — has long disappeared from spoken language. The domain of the preterite indicative (e.g. *hizo* "he did") — which likewise shared substantially the same feature values as the French simple past — has been progressively encroached upon by the present perfect indicative (e.g. *ha hecho* "he has done"), which, like the French compound past, has been "*derivando hacia el estadio número cuatro de aoristización*" (Montes 2019: 48). Interestingly, the narrative present in French first appears in the 12th(?) century, whereas in Spanish it does not emerge until the 16th(?) century. It is therefore highly likely that Spanish followed the same historical trajectory as French, albeit with a delay. Turning briefly to other Romance languages: in Italian — a tense-driven system — the contraction of the remote past and the expansion of the recent past are both particularly pronounced. By contrast, Galician and Asturian, which, like Portuguese, employ a perspective-and-aspect-driven temporal system, have retained a vital simple past, while the present perfect remains a marginal form.

5.3 The Marked Erosion of the Future Tense Forms in Spoken Portuguese

In Portuguese, future-tense forms are in clear decline, whereas in French, future situations are still generally expressed by the synthetic future, and Spanish appears to occupy an intermediate stage. In Portuguese, the combination of [$+VP \in \text{past}$] and [$-\text{finished}$] allows the present indicative to be used for referring to future situations. Moreover, the feature [$-VP \in \text{past}$] permits free displacement of the temporal viewpoint within the NON-PAST interval, which in turn makes it possible for the preterite indicative to be used with a future perfect interpretation. In other words, the language can express both futurity and future perfect meanings without recourse to the dedicated future-tense forms. In Portuguese, once the situation has been positioned in the future — via the specification [$-VP \in \text{past}$, $-\text{finished}$] and contextual cues such as adverbial phrases — the modal evaluation of its likelihood is secondarily marked by [$\pm \text{assumptive}$], as shown in (122) = (14). Thus, [$+\text{assumptive}$] functions solely to indicate the speaker's doxastic evaluation of the situation's likelihood, and plays no essential role in temporal anchoring.

Furthermore, since other markers of doxastic possibility are readily available (e.g. *se calhar*, *acho que*, *será que*), there is no necessity for [+assumptive] to be expressed morphologically. Taken together, these facts explain why the future tense in Portuguese has been functionally devalued both as a tense and as a modal marker, and therefore is undergoing systemic decline.

(122)

Temporal anchoring process				
	1 st stage	2 nd stage	3 rd stage	4 th stage
PT	[±VPEpast] speaker's temporal viewpoint located	[±finished] finishedness evaluated → situation time located	[±assumptive] occurrence possibility evaluated	
SP	[±past] reference times defined	[±posterior]←[±assumptive] sub-reference times defined	[±anterior] situation time located	[±VPEpast] viewpoint selected
FR	[±past] reference times defined	[±posterior] sub-reference times defined	[±anterior] situation time located	[±VPEpast] viewpoint selected

PT, SP, and FR stand for Portuguese, Spanish and French, respectively.

By contrast, in French and Spanish, the second stage of the temporal anchoring process requires, as indicated in (122), the establishment of a sub-reference time in order to locate the situation in the future. This is encoded either by the relative tense feature [+posterior] in French, or by the modal feature [+assumptive] in Spanish, which undergoes a metonymic extension to [+posterior] (on the metonymic extension from [+assumptive] to [+posterior], based on a conditional-consequential relation, see Section 2.2). Unlike in Portuguese, therefore, the feature [+assumptive] in these languages retains functional value as a necessary component of temporal anchoring process. In the case of Spanish, however, if the establishment of a future reference time via the metonymic extension [+assumptive] → [+posterior] is deemed unnecessary, then the synthetic future tense may face the risk of decline — as has already occurred in Portuguese.

Finally, when comparing French and Spanish — both of which employ tense-driven temporal anchoring systems — the reason why future reference is less easily expressed by the present indicative in French becomes clear. French encodes the absolute tense feature [±past], and the value [−past] is constrained to coincide with its default value, namely the speech time. As a result, forms specified as [−past], such as the present indicative and the compound past (*passé composé*), are structurally unsuited for expressing futurity. In such cases, future reference must be conveyed via [+posterior] forms, i.e., the simple future or the future perfect. The same applies to [+past] contexts: once a past reference time has been established, it cannot be shifted rightward. Accordingly, future-in-the-past constructions require the use of [+posterior] forms such as the past future or the past future perfect. By contrast, Spanish allows greater flexibility

in shifting the psychological present rightward along the time axis. Consequently, future reference can be expressed to some extent even by the present indicative.

5.4 The Future Perfect Use of the Preterite with the temporal adverbs *já/ainda não*

The preterite indicative in Portuguese can, as in (123), express a future perfect meaning, particularly in spoken language, when co-occurring with temporal adverbs such as *já* "already" or *ainda não* "not yet". (124) represents the corresponding form in the future perfect indicative:

(123) *Quando chegares, já ele morreu.* "When you arrive, he already died."

(124) *Quando chegares, ele (já) terá morrido.* "When you arrive, he will (already) have died."

This type of usage is possible because the Portuguese preterite indicative is not specified as [+past], but rather as [−VP∈past, +finished]. In (123), for example, the speaker's temporal viewpoint shifts from the speech time — the default value of the NON-PAST — to a future time, which is indicated by the adverbial clause *quando chegares*. From that future viewpoint, the situation is viewed as finished, which is encoded by the aspectual feature [+finished]. This future-perfect-like use of the Portuguese preterite indicative has, to our knowledge, no counterpart in the Latin perfect indicative. It is likely to be a Romance innovation. What made such an extension possible, however, is precisely the structural configuration of Portuguese's perspective-and-aspect-driven temporal anchoring system.

By contrast, neither the preterite indicative nor the present perfect in Spanish allows this kind of usage. Both (125) and (126) are ungrammatical. (127) represents the corresponding form in the future perfect indicative:

(125) **Cuando llegues, él yá murió.* "When you arrive, he already died."

(126) **Cuando llegues, él yá ha muerto.* "When you arrive, he has already died."

(127) *Cuando llegues, él (ya) habrá muerto.* "When you arrive, he will (already) have died."

French behaves in a similar manner: both (128) and (129) are ungrammatical. (130) represents the corresponding form in the future perfect indicative:

(128) **Quand tu arriveras, il mourut déjà.* "When you arrive, he already died."

(129) **Quand tu arriveras, il est déjà mort.* "When you arrive, he has already died."

(130) *Quand tu arriveras, il sera (déjà) mort.* "When you arrive, he will (already) have died."

In their tense-driven temporal anchoring systems, the Spanish preterite indicative and the French simple past are specified, respectively, as [+past, –assumptive, –anterior, –VP∈past] and [+past, –posterior, –anterior, –VP∈past]. These forms describe past situations [+past] as finished [+finished] relative to the speech time — the latter value being derived from [–VP∈past] —and therefore cannot be used to refer to future situations. Similarly, the Spanish present perfect and the French compound past are specified as [–past, –assumptive, +anterior] and [–past, –posterior, +anterior], respectively. These forms describe situations as finished [+finished], relative to the speech time [–past], with [+finished] derived metonymically in this case from [+anterior], and likewise cannot be used to refer to future situations. It should be noted, however, that in Spanish, if the feature [assumptive] becomes morphologically or functionally eroded, and if the temporal interval marked by [–past] extends rightward (i.e., into the near future), then it becomes theoretically possible for the present perfect to take on a future perfect reading. Before that, however, it is also theoretically possible for a construction involving the verb *ir* in the present indicative + *a* + perfect infinitive to be established for the expression of a future perfect reading — as is already the case in Brazilian Portuguese, where *ir* in the present indicative followed by a perfect infinitive functions as a future perfect. For example: *Quando você chegar, ele já vai ter morrido* ("When you arrive, he will already have died").

5.5 Cross-Romance Variation in Tense Selection in Counterfactual Constructions

The opposition between a perspective-and-aspect-driven temporal anchoring system and a tense-driven one also manifests itself in the temporal marking of counterfactual constructions. Consider the factual statement "Since I did not have money, I did not buy that car" and its corresponding counterfactual: "If I had had money, I would have bought that car":

(131) *Como não tinha dinheiro, não comprei aquele carro.* (factual)

(132) *Se eu tivesse dinheiro, tinha comprado aquele carro.* (counterfactual)

(133) *?Se eu tivesse tido dinheiro, tinha comprado aquele carro.* (counterfactual)

In spoken Portuguese, the counterfactual corresponding to the factual statement in (131) is given in (132), while (133) is typically judged to be unnatural. The contrast lies in the choice of verb form in the subordinate conditional clause: in this case, the simple form *tivesse* is preferred,

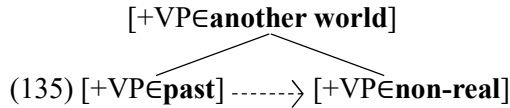
in parallel with the simple form *não tinha* in the factual statement, whereas the compound form *tivesse tido* is often perceived as pragmatically infelicitous. In this language, temporal anchoring proceeds in the order $[\pm VP \in \text{past}] \Rightarrow [\pm \text{finished}]$ as shown in (134) = (14), and anchoring cannot be effected unless both of these features are activated:

(134)

Temporal anchoring process				
	1 st stage	2 nd stage	3 rd stage	4 th stage
PT	$[\pm VP \in \text{past}]$ speaker's temporal viewpoint located	$[\pm \text{finished}]$ finishedness evaluated → situation time located	$[\pm \text{assumptive}]$ occurrence possibility evaluated	
SP	$[\pm \text{past}]$ reference times defined	$[\pm \text{posterior}] \leftarrow [\pm \text{assumptive}]$ sub-reference times defined	$[\pm \text{anterior}]$ situation time located	$[\pm VP \in \text{past}]$ viewpoint selected
FR	$[\pm \text{past}]$ reference times defined	$[\pm \text{posterior}]$ sub-reference times defined	$[\pm \text{anterior}]$ situation time located	$[\pm VP \in \text{past}]$ viewpoint selected

PT, SP, and FR stand for Portuguese, Spanish and French, respectively.

In counterfactual constructions, however, a metaphorical extension from $[+VP \in \text{past}]$ to $[+VP \in \text{non-real}]$ occurs via the schema *another world* or *intervention impossibility*, as shown in (135):



As a result, **it becomes impossible to distinguish whether the speaker's temporal viewpoint within the NON-REAL world is situated in the PAST or in the NON-PAST.** Moreover, even if one were to attempt to rely on the figurative extension of other features, such an approach would prove unworkable, since none of those features possesses an internal conceptual structure analogous to that of $[\pm VP \in \text{past}]$ — namely, the structure of "locating something somewhere". Accordingly, **whether the temporal viewpoint is situated in the PAST or in the NON-PAST must be inferred from contextual or situational cues.** By contrast, **the presence or absence of finishedness of a situation can still be explicitly encoded by the feature $[\pm \text{finished}]$.** On this basis, temporal anchoring in counterfactual constructions proceeds according to the sequence shown in (136):

(136) $[+VP \in \text{non-real}] (\leftarrow [+VP \in \text{past}]) \Rightarrow [\pm VP \in \text{past}]$ (determined by contextual or situational cues) $\Rightarrow [\pm \text{finished}]$

Returning to examples (131) – (133), the form *não tinha* in the factual sentence (131) is

specified as $[-\text{non-assertive}, +\text{VP}\in\text{past}, -\text{finished}]$. In order to convert this into a hypothetical statement — that is, one involving the suspension of truth-value judgment — within the NON-REAL domain, two operations are required, as shown in (137). First, the mood feature $[-\text{non-assertive}]$ of the indicative must be changed to $[+\text{non-assertive}]$ to yield the subjunctive form, and second, a metaphorical extension from $[+\text{VP}\in\text{past}]$ to $[+\text{VP}\in\text{non-real}]$ must be applied:

$$(137) [-\text{non-assertive}, +\text{VP}\in\text{past}, \boxed{-\text{finished}}] \rightarrow [+ \text{non-assertive}, +\text{VP}\in\text{non-real}, \boxed{-\text{finished}}]$$

When the verb *ter* bears this set of feature values, its surface realisation is the simple form *tivesse* as found in the counterfactual construction (132). In (137), the placement of the speaker's temporal viewpoint in the PAST — that is, the value $[+\text{VP}\in\text{past}]$ — must be inferred from contextual or situational cues. This is not difficult, however, given the presence of *tinha comprado* in the apodosis. Although its primitive specification is $[+\text{VP}\in\text{past}, +\text{finished}]$, it undergoes a metaphorical extension in this counterfactual construction to $[+\text{VP}\in\text{non-real}, +\text{finished}]$. Through the combination of this $[+\text{finished}]$ value with the default value of the speaker's temporal viewpoint — namely, the speech time encoded by $[-\text{VP}\in\text{past}]$ — the situation-base denoted by *tinha comprado* is derivatively anchored in the PAST, due to the composite feature configuration $[-\text{VP}\in\text{past}, +\text{finished}]$. The time of this situation-base's occurrence in the PAST then serves as the reference time at which the speaker's viewpoint is located in *tivesse*, which is therefore interpreted as $[+\text{VP}\in\text{past}]$. This $[+\text{VP}\in\text{past}]$ value, when combined with the $[-\text{finished}]$ value inherently encoded by *tivesse*, yields the composite feature configuration $[+\text{VP}\in\text{past}, -\text{finished}]$. As a result, the state *eu ter dinheiro* is interpreted as temporally simultaneous with the past event *eu comprar aquele carro*, and as the condition that could have enabled it.

The compound form *tivesse tido* in (133), by contrast, becomes specified as $[+\text{non-assertive}, +\text{VP}\in\text{non-real}, +\text{finished}]$ via the same metaphorical extension. The $[+\text{finished}]$ value then combines with the contextually generated $[+\text{VP}\in\text{past}]$ value — a feature that encodes the reference time simultaneous with *tinha comprado*, specified as $[-\text{VP}\in\text{past}, +\text{finished}]$, and at which the speaker's temporal viewpoint is located — yielding the feature combination $[+\text{VP}\in\text{past}, +\text{finished}]$. As a result, the state *eu ter dinheiro* is interpreted as already finished — that is, as no longer existing — at the time of the past event *eu comprar aquele carro* and thus fails to be interpreted as a viable condition that could have enabled that event.

Now, let us suppose that Portuguese did not employ $[\pm\text{VP}\in\text{past}] \times [\pm\text{finished}]$ for temporal anchoring, but instead used the combination of reference time and aspectual features $[\pm\text{past}] \times [\pm\text{finished}]$. Under such a system, the following would occur: in counterfactual constructions,

the feature [+past] would undergo a metaphorical extension of the (132) type to [+non-real], in order to signal non-reality; as a result, the feature [±past] could no longer be used to represent the reference time within the thus created counterfactual world; it would become possible, however, to recover a reference time specification through a chain of semantic extensions: the aspectual feature [±finished] (cause) would first be metonymically extended to the situation time feature [±anterior] (effect), and then this [±anterior] feature would undergo a further metaphorical extension to [±past], now reinterpreted as a reference time feature; as a consequence, [±finished] would lose its original aspectual value, and because no feature would remain to explicitly represent it, the [+finished]/[−finished] opposition would instead have to be inferred from context or situational cues; the temporal anchoring process in counterfactual constructions under such a system would therefore proceed as schematised in (138):

$$(138) \text{ [+non-real] } (\leftarrow \text{ [+past]}) \Rightarrow \text{ [±past] } (\leftarrow \text{ [±anterior] } \leftarrow \text{ [±finished]}) \Rightarrow \text{ [±finished]} \\ \text{(determined by contextual or situational cues)}$$

In this case, the counterfactual counterpart to the factual sentence (131) would not be (132), but rather (133). The verb *não tinha* in the factual clause is specified as [−non-assertive, +past, −finished], and when transferred into the counterfactual domain, it would assume the value [+non-assertive, +non-real, +past, (−finished)]. The only form capable of carrying this specification would be *tivesse tido*, whose primitive feature configuration would be [+non-assertive, +past, +finished]. In the counterfactual construction, the following series of figurative extensions in (139) would take place:

$$(139) \text{ [+non-assertive, +past, +finished]} \Rightarrow \text{ [+non-assertive, +non-real, +past]}$$

The [−finished] value of *não tinha* should, in the case of *tivesse tido*, be inferred from context or situational cues. As will be discussed in Section 5.7, **Classical Latin is precisely a language of this type**. Conversely, the fact that **the feature [±finished] retains its original aspectual meaning in Portuguese counterfactual constructions** constitutes strong evidence that **Portuguese does not employ [±past] × [±finished], but rather [±VPEpast] × [±finished] as the basis for temporal anchoring**.

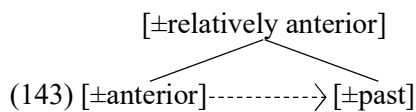
By contrast, in Spanish, the counterfactual sentence corresponding to the factual one in (140) is (142), whereas (141) is judged to be ungrammatical. In this case — in contrast to Portuguese — the verb form that corresponds to the simple form *tenía* in the factual sentence (140) is the compound form *hubiera tenido* in (142), rather than the simple form *tuviera* in (141). What does this imply?

(140) *Como yo no tenía dinero, no compré aquel coche.* (factual)

(141) **Si yo tuviera dinero, habría comprado aquel coche.* (counterfactual)

(142) *Si yo hubiera tenido dinero, habría comprado aquel coche.* (counterfactual)

In this language, temporal anchoring proceeds in the sequence $[\pm\text{past}] \Rightarrow [\pm\text{posterior}] (\leftarrow [\pm\text{assumptive}]) \Rightarrow [\pm\text{anterior}]$ as shown in (134) above, and anchoring cannot begin unless the absolute tense feature $[\pm\text{past}]$ is first activated. In Spanish, however, a metaphorical extension from $[\pm\text{past}]$ to $[\pm\text{non-real}]$ occurs in counterfactual constructions via the schema *another world* or *intervention impossibility*, as in (135). As a result, **within the NON-REAL world thus construed, it is no longer possible to use the feature $[\pm\text{past}]$ to indicate whether the situation precedes the speech time — encoded as $[\pm\text{past}]$ — or not — encoded as $[\pm\text{past}]$.** At the same time, since $[\pm\text{past}]$ constitutes the first stage of temporal anchoring, it must still be realised morphologically. To satisfy this requirement, a metaphorical extension arises from $[\pm\text{anterior}]$ to $[\pm\text{past}]$ via the schema of *relatively anterior* — that is, "anterior to some temporal point", whether it is the speech time or a reference time marked as $[\pm\text{past}, \pm\text{posterior}]$:



This figurative extension in (143) makes temporal anchoring possible. Accordingly, temporal anchoring in Spanish counterfactual constructions proceeds — unlike in Portuguese, as shown in (136) — according to the sequence in (144):

(144) $[\pm\text{non-real}] (\leftarrow [\pm\text{past}]) \Rightarrow [\pm\text{past}] (\leftarrow [\pm\text{posterior}]) \Rightarrow [\pm\text{anterior}]$ (determined by contextual or situational cues)

Returning to examples (140) – (142), the form *no tenía* in the factual sentence (140) is specified as $[-\text{non-assertive}, +\text{past}, -\text{assumptive}, -\text{anterior}, -\text{VP} \in \text{past}]$. To convert this into the expression of a hypothetical situation situated within the NON-REAL domain, the construction must be mapped onto a form capable of bearing the feature values $[\pm\text{non-assertive}, +\text{non-real}, +\text{past}, -\text{anterior}]$, as required in counterfactual contexts. (It should be noted that within the domain marked as $[\pm\text{non-assertive}]$, the oppositions $[\pm\text{assumptive}]$ and $[\pm\text{VP} \in \text{past}]$ are neutralised.) The only form that satisfies these conditions is *hubiera tenido* in (142), marked as $[\pm\text{non-assertive}, +\text{past}, +\text{anterior}]$. This is because, as shown in (145), this form allows for the feature extension in (143):

(145) *hubiera tenido* $[\pm\text{non-assertive}, +\text{past}, \boxed{+\text{anterior}}] \rightarrow [\pm\text{non-assertive}, +\text{non-real}, \boxed{+\text{past}}]$

The right-hand side feature set [+non-real, +past] arises through a semantic extension from the left-hand side specification [+past, +anterior]. In this extension, [+anterior] is metaphorically extended to [+past] as shown in (143), which makes it impossible to employ this feature in order to express the original [−anterior] value associated with *tenía*: it is thus from contextual or situational cues that one infers whether the situation time is to be understood as anterior to, or simultaneous with, the reference time marked as [+past]. By contrast, in the case of *tuviera* in (141), marked as [+past, −anterior, +non-assertive], a different extension occurs, as shown in (146):

(146) *tuviera* [+non-assertive, +past, −anterior] → [+non-assertive, +non-real, −past]

This form is consequently ruled out, since it cannot encode the [+past] value that was present in *tenía*. These differences in tense selection in counterfactual constructions between Portuguese and Spanish arise entirely from the structural opposition between the two temporal anchoring systems. The Spanish-type treatment also holds in other languages that employ a tense-driven temporal anchoring system, such as French, Italian and English, as shown in (147) – (155):

(147) *Comme je n'avais pas d'argent, je n'ai pas acheté cette voiture-là.* (factual)

(148) **Si j'avais de l'argent, j'aurais acheté cette voiture-là.* (counterfactual)

(149) *Si j'avais eu de l'argent, j'aurais acheté cette voiture-là.* (counterfactual)

(150) *Poiché non avevo i soldi, non ho comprato quella macchina.* (factual)

(151) **Se avessi i soldi, avrei comprato quella macchina.* (counterfactual)

(152) *Se avessi avuto i soldi, avrei comprato quella macchina.* (counterfactual)

(153) *Since I did not have money, I did not buy that car.* (factual)

(154) **If I had money, I would have bought that car.* (counterfactual)

(155) *If I had had money, I would have bought that car.* (counterfactual)

5.6 The maintenance of the preterite-imperfect opposition in the subjunctive in Portuguese and its neutralisation in Spanish and French

In contemporary standard spoken and written Spanish, the speaker's temporal perspective — that is, the feature [±VP∈past] — is morphologically encoded only in the opposition between

the preterite and imperfect indicatives, as shown by the two shaded cells in (156) = (9). This feature is no longer morphologically expressed in the pluperfect indicative (e.g. *había hablado*), which once stood in both morphological and semantic opposition to the now nearly obsolete preterite anterior (e.g. *hubo hablado*):

(156)	[−assumptive]		[+assumptive]		
[−past]	hablo	ha hablado	hablará	habrá hablado	[−non-assertive]
[+past]	[+VP∈past] hablaba	había hablado	hablaría	habría hablado	
	[−VP∈past] habló				
[−past]	hable	haya hablado	_____	_____	[+non-assertive]
[+past]	hablara	hubiera hablado	_____	_____	
	[−anterior]	[+anterior]	[−anterior]	[+anterior]	

Today, apart from these two forms, there exists no pair of finite verb forms that are identical in all values of the features [±non-assertive], [±past], [±assumptive], and [±anterior]. Accordingly, the speaker's temporal perspective is not morphologically encoded elsewhere in the system; its interpretation is context-situation-dependent — or possibly determined by some default setting (a possibility that merits further investigation).

The imperfect indicative and the preterite indicative, which are both specified as [−non-assertive, +past, −assumptive, −anterior], differ solely in terms of the speaker's temporal viewpoint. In the imperfect indicative, this viewpoint is located at the reference time, which is specified as [+past, −assumptive] — effectively [+past]. Since the situation time is [−anterior] — that is, simultaneous with that reference time — the speaker places their viewpoint within the time interval along which the past situation was developing, and describes it from within that temporal stretch.

By contrast, in the preterite indicative, the speaker's temporal viewpoint is located differently. Given the structure of reference time in Spanish, there are three logical candidates: [+past, +assumptive], [−past, −assumptive], and [−past, +assumptive]. Of these, [+past, +assumptive] corresponds to a past-future reference time, and [−past, +assumptive] to a future reference time. In either of these cases, the past situation would be described from an artificially displaced viewpoint, the motivation for which would be difficult to explain. The remaining candidate, [−past, −assumptive], is effectively [−past] — that is, the speech time — and serves as the default location for the speaker's temporal viewpoint. It is thus most natural for the speaker to describe the past situation from this present-time viewpoint — that is, from a viewpoint external

to the temporal domain in which the situation occurred.

In contemporary standard Spanish, then, the preterite indicative is the only tense that explicitly marks the speaker's temporal viewpoint as located at the speech time — that is, in the present — while situating the situation to be described in the past. As such, it possesses a fundamentally distinct feature structure from other tense forms used to describe past situations. This separation between the speaker's perspective and the situation time appears to be the key to explaining the preterite's implication of not only finishedness but also perfectivity, verifiability, and factual presentation — that is, objectivity in general. The fact that the French preterite indicative — *passé simple* — which shares the same feature configuration, disappeared very early from spoken usage may be seen as the logical consequence of this "cold" objective mode of anchoring past situations.

As seen above and in Section 4, the opposition of perspective between $[-VP \in \text{past}]$ and $[+VP \in \text{past}]$ is only attested within the domain of the indicative forms specified as $[-\text{non-assertive}, +\text{past}, -\text{assumptive}, -\text{anterior}]$; it is neutralised in that of the subjunctive forms specified as $[+\text{non-assertive}, +\text{past}, -\text{assumptive}, -\text{anterior}]$. As a result, in the subjunctive mood, no contrast is found between the preterite with $[-VP \in \text{past}]$ and the imperfect with $[+VP \in \text{past}]$.

There appear to be three reasons for this neutralisation. First, as previously stated, the perspective feature $[\pm VP \in \text{past}]$ only becomes active at the fourth stage of the temporal anchoring process described in (157) = (14). However, the temporal localisation of the situation itself has already been completed at the preceding third stage, before this feature becomes active. Accordingly, the functional value of this feature for temporal anchoring is relatively low:

(157)

Temporal anchoring process				
	1 st stage	2 nd stage	3 rd stage	4 th stage
PT	$[\pm VP \in \text{past}]$ speaker's temporal viewpoint located	$[\pm \text{finished}]$ finishedness evaluated → situation time located	$[\pm \text{assumptive}]$ occurrence possibility evaluated	
SP	$[\pm \text{past}]$ reference times defined	$[\pm \text{posterior}] \leftarrow [\pm \text{assumptive}]$ sub-reference times defined	$[\pm \text{anterior}]$ situation time located	$[\pm VP \in \text{past}]$ viewpoint selected
FR	$[\pm \text{past}]$ reference times defined	$[\pm \text{posterior}]$ sub-reference times defined	$[\pm \text{anterior}]$ situation time located	$[\pm VP \in \text{past}]$ viewpoint selected

PT, SP, and FR stand for Portuguese, Spanish and French, respectively.

Second, in contemporary usage, within the indicative mood — that is, $[-\text{non-assertive}]$ — the

only semantic opposition generated by this feature is the one between the preterite and the imperfect, and only within the narrow domain [−non-assertive, +past, −assumptive, −anterior]. Consequently, this feature exhibits low functional efficiency in generating oppositions, while incurring a high maintenance cost as a highly marked feature.

Third, as is evident from (158), (159), (160) below = (5), (9), (12), in all three languages — Portuguese, Spanish and French — the gaps in the system of temporal-reference forms (highlighted in shading) consistently occur within the domain of the subjunctive. This suggests that the subjunctive mood itself, marked as [+non-assertive], is already a marked category in contrast to the unmarked indicative, and that in Spanish, it tends to resist co-occurrence with another marked feature such as [±VP∈past]:

(158)	[−finished]	[+finished]	[−finished]	[+finished]	
[−VP∈past]	[0 initiated, 0 stative] fala	falou	falará	terá falado	[−non-assertive]
	[✓initiated, ✓stative] tem falado				
[+VP∈past]	falava	tinha falado	falaria	teria falado	[+non-assertive]
[−VP∈past]	fale	tenha falado	falar	tiver falado	
[+VP∈past]	falasse	tivesse falado	————	————	
	[−assumptive]		[+assumptive]		

(159)	[−assumptive]		[+assumptive]		
[−past]	hablo	ha hablado	hablará	habrá hablado	[−non-assertive]
[+past]	[+VP∈past] hablaba	había hablado	hablaría	habría hablado	
	[−VP∈past] habló				
[−past]	hable	haya hablado	————	————	[+non-assertive]
[+past]	hablara	hubiera hablado	————	————	
	[−anterior]	[+anterior]	[−anterior]	[+anterior]	

(160)	[−posterior]		[+posterior]		
[−past]	il parle	il a parlé	il parlera	il aura parlé	[−non-assertive]
[+past]	[+VP∈past] il parlait	[+VP∈past] il avait parlé	il parlerait	il aurait parlé	
	[−VP∈past] il parla	[−VP∈past] il eut parlé			
[−past]	il parle	il ait parlé	————	————	[+non-assertive]
[+past]	il parlât	il eût parlé	————	————	
	[−anterior]	[+anterior]	[−anterior]	[+anterior]	

As seen in (157) = (14), the same held true in literary written French, which operated with the same tense-driven system of temporal anchoring as Spanish. In the indicative, specified as [−non-assertive], the opposition between [−VP∈past] and [+VP∈past] — that is, the opposition

between the preterite *il parla* and the imperfect *il parlait*, on the one hand, and between the pluperfect *il avait parlé* and the preterite anterior *il eut parlé*, on the other — was neutralised in the subjunctive, specified as [+non-assertive]. The imperfect subjunctive *il parlât* and the pluperfect subjunctive *il eût parlé* were neutralised counterparts, reflecting the same explanatory factors discussed above. Examples (161), (162) and (163), which feature the preterite indicative *il sut*, the imperfect indicative *il savait*, and the imperfect subjunctive *il sût* of the verb *savoir* "to know" in the third person singular, illustrate this neutralisation:

(161) *Il sut la vérité.* "He found out the truth." [–non-assertive, –VP€past]

(162) *Il savait la vérité.* "He knew the truth." [–non-assertive, +VP€past]

(163) *Je ne crois pas qu'il sût la vérité.* "I don't believe he found out / knew the truth." [+non-assertive, 0 VP€past]

By contrast, Portuguese operates with a perspective-and-aspect-driven temporal anchoring system, in which, as shown in (157) = (14), temporal anchoring occurs only after the process [±VP€past] ⇒ [±finished] has been completed. Accordingly, unless both the perspective feature [±VP€past] and the aspectual feature [±finished] are activated, the temporal anchoring of the situation cannot be achieved. For this reason, even in the subjunctive mood — i.e., in the [+non-assertive] domain — the expression of both [±VP€past] and [±finished] is obligatory, just as it is in the indicative mood, specified as [–non-assertive]. This obligatory expression is formally manifested in the opposition between the preterite subjunctive *tenha falado* — [+non-assertive, –VP€past, +finished] — and the imperfect subjunctive *falasse* — [+non-assertive, +VP€past, –finished].

5.7 The systemic commonalities and differences in temporal anchoring between Classical Latin and Portuguese

First, Classical Latin — henceforce, Latin — possessed an opposition between the indicative and subjunctive moods, which can be specified as [–non-assertive] and [+non-assertive], respectively — just as Portuguese, Spanish and French do — as illustrated in (164) and (165):

(164) *Sī hoc dīcis, errās.* "If you say this, you are mistaken." (present indicative)

(165) *Sī hoc dīcās, errēs.* "If you should say this, you would be mistaken." (present subjunctive)

Second, as stated in Sections 0 and 5.1, the Portuguese *Pretérito Perfeito do Indicativo* — PPS — equals the Latin *Perfectum Indicativi* in that both encode the [+finished] value, as exemplified in (166) and (167), and the feature [+finished] is the key concept in the semantic reanalysis of the PPC:

(166) *Hodiē amīcus mihi librum dōnāvīt.* "Today a friend gave me a book."

(167) *Hoje um amigo deu-me um livro.* "Idem."

Nevertheless, the core temporal-aspectual system of Latin is not completely identical to that of Modern Portuguese, which is defined by a 2×2 feature grid of [±VP∈past] × [±finished].

In Latin, the imperfect subjunctive — e.g. *amārem, amārēs, amāret, amārēmus, amārētis, amārent* from *amāre* — and the pluperfect subjunctive — e.g. *amāvissem, amāvissēs, amāvisset, amāvissēmus, amāvissētis, amāvissem* from the same verb — were used to express counterfactuals in the non-past and in the past, respectively, as shown in (168) and (169):

(168) *Sī hoc dīcerēs, errārēs.* "If you said this, you would be mistaken."

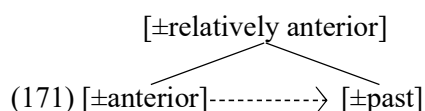
(169) *Sī hoc dīxissēs, errāvissēs.* "If you had said this, you would have been mistaken."

The feature compositions of the imperfect and pluperfect subjunctives can be assumed to have been, respectively, [+non-assertive, +past, −finished] and [+non-assertive, +past, +finished] or, alternatively, [+non-assertive, +VP∈past, −finished] and [+non-assertive, +VP∈past, +finished]. The value of the futurity-relevant feature [posterior] or [assumptive] — to be discussed later — was not specified in these forms, since neither a past future subjunctive nor a past future perfect subjunctive existed.

In counterfactuals, the feature [+past] or [+VP∈past], as assumed above, had to undergo a figurative extension into [+non-real] or [+VP∈non-real] in order to denote counterfactuality, and the remaining [±finished] then had to undergo an extension into [±past] or [±VP∈past]. However, since [±finished] does not possess an internal conceptual structure of "locating something somewhere" characteristic of [±VP∈past] (cf. Section 5.5), the shift [±finished] → [±VP∈past] would be impossible. Consequently, the extension [±finished] → [±past] took place. In this semantic shift, it is plausible to assume the occurrence of a metonymy in the reverse direction of (103) — "metonymy: [+anterior] (effect) → [+finished] (cause)" — as indicated in (170):

(170) metonymy: [±finished] (cause) → [±anterior] (effect)

Then, metaphorical extension arises from $[\pm\text{anterior}]$ to $[\pm\text{past}]$ via the schema of *relatively anterior* — that is, "anterior to some temporal point", whether it is the speech time or a reference time marked as $[\pm\text{past}]$, as shown in (171) = (143):



From the above discussion, **the core temporal-aspectual system of Latin can be defined** not by $[\pm\text{VP}\in\text{past}] \times [\pm\text{finished}]$, but rather **by a 2×2 feature grid of $[\pm\text{past}] \times [\pm\text{finished}]$** . By contrast, in Portuguese, as discussed in Section 5.5, $[\pm\text{VP}\in\text{past}]$ in counterfactuals undergoes a metaphorical extension into $[\pm\text{VP}\in\text{non-real}]$, while the remaining feature $[\pm\text{finished}]$, owing to its conceptual-structural difference from $[\pm\text{VP}\in\text{past}]$, cannot be extended into the latter and therefore remains as $[\pm\text{finished}]$; $[\pm\text{VP}\in\text{past}]$, in turn, must be inferred from contextual or situational cues. This provides clear evidence that **Portuguese employs $[\pm\text{VP}\in\text{past}]$ rather than $[\pm\text{past}]$** . It therefore follows that the Latin system of $[\pm\text{past}] \times [\pm\text{finished}]$ shifted to that of $[\pm\text{VP}\in\text{past}] \times [\pm\text{finished}]$ in Portuguese, although the question of when and how this shift occurred is left for future research. It should be noted, in this respect, that $[\pm\text{past}, \pm\text{finished}]$, for example, merely indicates the positional relation between the past reference time and the end point of the situation time — in this case, $[\pm\text{finished}]$ encodes that the end point is anterior to the past reference time — and does not, unlike $[\pm\text{VP}\in\text{past}, \pm\text{finished}]$, imply that the speaker's VP is explicitly located at that past time and, moreover, can be displaced along the time axis.

Third, another major difference between Portuguese and Latin is that the latter lacked the past future and past future perfect indicatives as well as the future and future perfect subjunctives. Moreover, the only future tenses in Latin — the future and future perfect indicatives — were, in contrast to those in Modern Portuguese, more temporal than modal. The future and future perfect indicatives are therefore assumed to have been specified as $[\pm\text{posterior}]$, which could metonymically be extended to $[\pm\text{assumptive}]$, depending on context and/or situation.

Summing up, temporal anchoring in Latin proceeded as follows. Its system first determined whether the situation to be described belonged to the actual world or to a non-actual one — a distinction encoded by **the mood feature $[\pm\text{non-assertive}]$** . Next, starting from the speech time as the primary anchoring point, two major reference times were established relative to it: the "past", which preceded the speech time, and the "non-past", which coincided with it. These two reference times were linguistically encoded by **the absolute tense feature $[\pm\text{past}]$** .

In the domain of the indicative specified as [−non-assertive], relative to the latter of these two primary reference times — namely, [−past] — two secondary reference times were then defined: the so called "future", which was posterior to the speech time as the primary reference time, and the so called "present", which was simultaneous with it. This opposition was encoded by **the relative tense feature [±posterior]**, which, when combined with [+past], assumed the value [0posterior], due to the absence of . In the domain of the subjunctive specified as [+non-assertive], by contrast, both primary reference times — namely, [+past] and [−past] — combine with [0posterior], since the subjunctive lacked any future tense.

With reference to one of the three reference-time combinations established in the indicative — namely, "present" [−past, −posterior], "future" [−past, +posterior] and "past" [+past, 0posterior] — and to one of the two reference-time combinations established in the subjunctive — namely, "present" [−past, 0posterior] and "past" [+past, 0posterior] — the end point of the situation time was evaluated as being either prior to it — that is, "perfect" — or simultaneous with it — that is, "non-perfect". This evaluation was encoded by **the aspectual feature [±finished]**.

The commonality and difference in the temporal anchoring process between Latin and Portuguese can be summarised as in (172):

(172)

Temporal anchoring process				
	1 st stage	2 nd stage	3 rd stage	4 th stage
LT	[±past] reference times defined	[±posterior] only with [−non-assertive, −past] sub-reference times defined	[±finished] finishedness evaluated → situation time located	
PT	[±VPEpast] speaker's temporal viewpoint located	[±finished] finishedness evaluated → situation time located	[±assumptive] occurrence possibility evaluated	

LT and PT stand for Classical Latin and Portuguese, respectively.

The most important commonality is that in both systems, as indicated by the shaded cells in (172), **the final stage of temporal anchoring process is executed by [±finished]**. Another commonality is that, **except for the domain [−non-assertive, −past] in Latin, the temporal anchoring process is completed in two stages**.

The Latin finite verb forms of the indicative and subjunctive and the Portuguese corresponding forms are exemplified in (173) and (174), respectively, using the third-person singular forms of *amāre* and *amar*. Forms that stand in an etymon-descendant relation and that have undergone

little change in their function of temporal anchoring are shaded:

(173)	[−finished]	[+finished]	[−finished]	[+finished]	
[−past]	amat	amāvit	amābit	amāverit	[−non-assertive]
[+past]	amābat	amāverat	————	————	
[−past]	amet	amāverit	————	————	[+non-assertive]
[+past]	amāret	amāvisset	————	————	
	[−posterior]		[+posterior]		

(174)	[−finished]	[+finished]	[−finished]	[+finished]	
[−VP∈past]	ama	amou	amará	terá amado	[−non-assertive]
[+VP∈past]	amava	amara tinha amado	amaria	teria amado	
[−VP∈past]	ame	tenha amado	amar	tiver amado	[+non-assertive]
[+VP∈past]	amasse	tivesse amado	————	————	
	[−assumptive]		[+assumptive]		

From (173) and (174), it is evident that **the Portuguese core temporal-aspectual system constituted by the shaded four non-future indicative forms** — namely, *ama*, *amou*, *amava*, *amara/tinha amado* — **has undergone virtually no functional change in temporal anchoring since the time of Latin, whose system was constituted by the shaded etymons of these four Portuguese indicative forms** — namely, *amat*, *amāvit*, *amābat*, *amā(ve)rat*. The only formal difference is that the Portuguese pluperfect indicative *amara*, which derives from the Latin pluperfect indicative *amā(ve)rat*, has been replaced in everyday usage of the modern language by the compound form *tinha amado*. The reason for this formal replacement has already been stated in Section 2.1.2.

Excluding these four indicative forms, the only one that has undergone virtually no such change is the Portuguese present subjunctive *ame*, which derives from the Latin present subjunctive *amet*. The Latin pluperfect subjunctive *amā(vi)ssem* and the Latin future perfect indicative *amā(ve)rit* have been shifted into the Portuguese imperfect subjunctive *amasse* and the Portuguese future subjunctive *amar*, respectively. All the remaining forms — namely, *tenha amado*, *tivesse amado*, *amará*, *terá amado*, *amaria*, *teria amado*, *tiver amado* — including *tinha amado*, are Portuguese creations.

6. Conclusion

This study has accounted for various cross-linguistic differences in temporal anchoring among Portuguese, other Romance languages — namely, Spanish and French — and Latin by means of three analytical components:

- **Set of primitive anchoring-aspect features**
- **Ordered sequence in which these features apply**
- **Metaphorical semantic extensions to which they are subject**

The temporal anchoring process in these languages can be schematised as illustrated in (175):

(175)

Temporal anchoring process				
	1 st stage	2 nd stage	3 rd stage	4 th stage
LT	[±past] reference times defined	[±posterior] only with [−non-assertive, −past] sub-reference times defined	[±finished] finishedness evaluated → situation time located	
PT	[±VPEpast] speaker's temporal viewpoint located	[±finished] finishedness evaluated → situation time located	[±assumptive] occurrence possibility evaluated	
SP	[±past] reference times defined	[±posterior]←[±assumptive] sub-reference times defined	[±anterior] situation time located	[±VPEpast] viewpoint selected
FR	[±past] reference times defined	[±posterior] sub-reference times defined	[±anterior] situation time located	[±VPEpast] viewpoint selected

LT, PT, SP and FR stand for Classical Latin, Portuguese, Spanish and French, respectively.

Depending on the minimal set of features employed for temporal anchoring and their order of application, at least three distinct systems of temporal anchoring can be distinguished within the Romance languages and Classical Latin:

- **Tense-and-aspect-driven** system — as found in Classical Latin
- **Perspective-and-aspect-driven** system — as found in Portuguese
- **Tense-driven** system — as found in Spanish and French

In the first system, temporal anchoring is achieved by constructing a (complex) reference time with respect to the speech time and locating the end point of the situation time relative to that reference time, following a sequence such as: [±past] ⇒ [±posterior] only with [−non-assertive, −past] ⇒ [±finished].

In the second, temporal anchoring is achieved by dividing the time axis into two intervals with respect to the speech time and determining whether the situation to be described is viewed as finished or not finished from a viewpoint situated within either interval: $[\pm VP \in \text{past}] \Rightarrow [\pm \text{finished}]$.

In the third, temporal anchoring is achieved by constructing a complex reference time with respect to the speech time and locating the situation to be described relative to that reference time, following a sequence such as: $[\pm \text{past}] \Rightarrow [\pm \text{posterior}] \Rightarrow [\pm \text{anterior}] \Rightarrow [\pm VP \in \text{non-past}]$ only with $[-\text{non-assertive}, +\text{past}, -\text{posterior}/-\text{assumptive}]$.

The cross-linguistic differences accounted for building on these systemic differences are:

- **Semantic reanalysis of the Portuguese present perfect indicative (PPC)**
- **Elimination of the preterite in French and aoristic shift of the present perfect in Spanish**
- **Decline of future tense forms and associated future-perfect use of the preterite in Portuguese**
- **Divergent selection patterns of temporal-aspectual forms in counterfactual constructions across Romance languages**
- **Maintenance of the preterite-imperfect opposition in the subjunctive in Portuguese and its neutralisation in Spanish and French**

Among these phenomena, particular emphasis has been placed on the reanalysis of the PPC, which represents — we want to think — a distinctive contribution of this study.

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